

University *of* Victoria Department *of* Mechanical Engineering

Malahat Express
Phase 2: Final Design

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Prepared For:

MECH 441: Ship Hydrostatics and Dynamics

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Summary

This document presents the Phase 2 design development of the Malahat Express, a passenger ferry connecting Brentwood Bay to Cowichan Bay as a marine bypass of the Malahat (Highway 1). Environmental design basis, principal dimensions, and preliminary hull form were established in Phase 1, where this phase advances the design through detailed machinery and tank arrangement, weight distribution, and stability analysis. Propulsion, Ro-Ro space, ballast, and liquid cargo systems are defined and placed within the hull, followed by the development of loading conditions incorporating lightship, deadweight, and payload estimates. Floodable length and intact stability assessments are then presented, verifying the vessel's compliance with intact stability criteria and establishing its freeboard values. Together, this covers the required Phase 2 information, completing the finalized design.

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1. Introduction and Background

1.1 Vessel Mission Statement

The mission of the Malahat Express is to provide a scheduled vehicle and passenger ferry service between Brentwood Bay on the Saanich Peninsula and Cowichan Bay in the Cowichan Valley, serving as a marine bypass of the Malahat section of the Trans-Canada Highway (Highway 1). The vessel shall operate year-round in all weather conditions typical of Saanich Inlet and Satellite Channel, providing a safe, reliable, and efficient alternative to highway travel for commuters, commercial vehicles, and recreational travellers.

1.2 Intended Route

The proposed route operates between the existing Brentwood Bay ferry terminal on the Saanich Peninsula and a new terminal at Cowichan Bay, transiting north through Saanich Inlet and Satellite Channel. Locating the southern terminus at the established Brentwood Bay site leverages existing road access and marine approaches, avoiding the need for an entirely new southern terminal, though berth modifications would be required to accommodate the larger vessel. Figure 1 shows the departure and return tracks over the approximately 18 km crossing. The coordinates of the Mill Bay Ferry Terminal in Brentwood Bay are 48.577°N, 123.467°W and the approximate coordinates of the new proposed Cowichan Terminal are 48.743°N, 123.630°W.

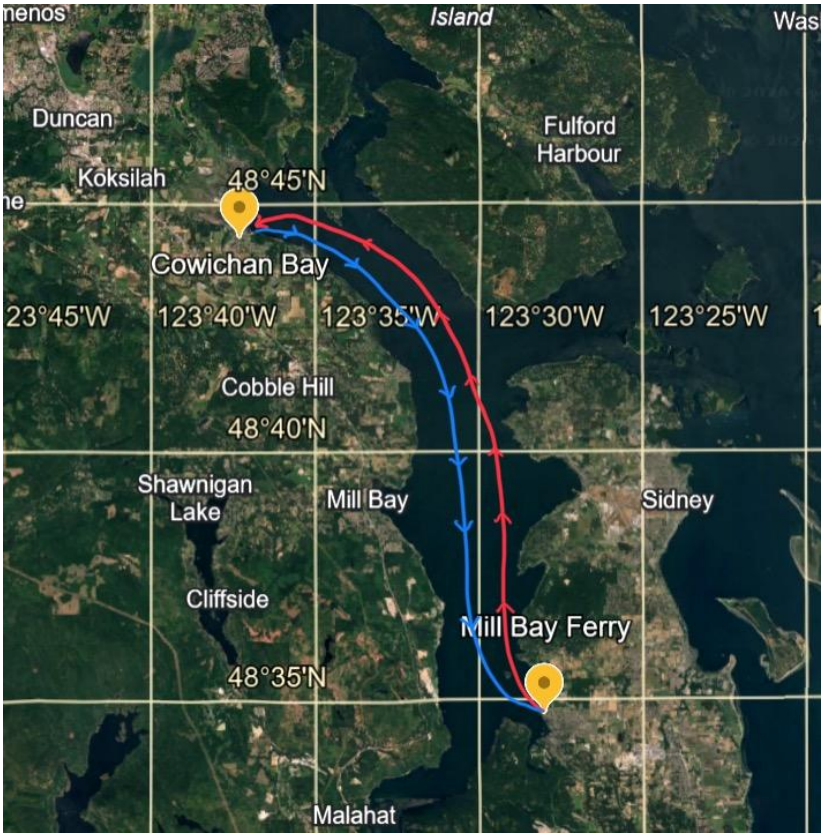


Figure 1: Intended Ferry Route

1.3 Operational Profile

The vessel is a short-haul commuter and vehicle ferry operating a fixed, high-frequency schedule, and its operational profile reflects a repetitive daily cycle rather than long ocean passages. Each one-way crossing of approximately 18 km takes about 40 minutes at the 14.5-knot service speed, and terminal turnaround is limited to 20 minutes at each berth. A single round-trip cycle therefore comprises roughly 80 minutes underway and 40 minutes at berth, so during active service, the vessel spends approximately two-thirds of its time transiting and one-third loading and unloading at the terminals. The transit route through Saanich Inlet and Satellite Channel is not subject to a scheduled speed restriction under the Vessel Operation Restriction Regulations [17]; open-water speed is governed by the COLREGs safe-speed requirement [18]. Near-shore and terminal-approach operations observe the 10 km/h (5.4 kn) within-30 m-of-shore convention. The Malahat Express complies with these requirements by design. Its 14.5 kn service speed applies only to the open-water portion of the crossing, where no scheduled limit applies and speed is set by the master in accordance with the safe-speed duty, accounting for visibility, traffic density, and sea state. On approach to the Brentwood Bay and Cowichan Bay terminals the vessel reduces to maneuvering speed, remaining below 5.4 kn within 30 m of shore, consistent with the near-shore convention.

With a minimum of eight round-trip sailings per day and increased frequency during the morning (06:00–09:00) and afternoon (15:00–19:00) commuter peaks, the vessel is in active service for an extended daily operating window, with the remaining hours spent idle at berth overnight. Averaged over the vessel's design life, and accounting for overnight lay-up, scheduled maintenance, and periodic dry-docking, the vessel is estimated to spend on the order of 45–55% of its total life underway, with the balance at berth or out of service.

Since the route lies entirely within the sheltered waters of Saanich Inlet and the more exposed Satellite Channel, the vessel encounters the most significant wave action during the transit portion of its cycle, particularly in Satellite Channel. The high proportion of operating time spent underway, combined with the year-round schedule, means the hull and structure are subject to frequent, repeated wave loading over short passages rather than occasional severe events; a fatigue-driven loading pattern characteristic of commuter ferries. This informs the environmental design conditions established in Section 1.5.

1.4 Environmental Conditions

The environment that the proposed vessel will move through are the Saanich Inlet, as well as Satellite Channel. The Saanich Inlet is a body of water that runs north-south between the Saanich Peninsula and the Malahat highlands, while Satellite Channel is an open channel at the Northern part of Saanich Inlet, separating the Saanich peninsula from Salt Spring Island. While the Saanich Inlet is topographically sheltered, the Satellite Channel is open to southeasterly and northeasterly wind as well as waves from the Strait of Georgia, which needs to be taken into consideration when designing for environmental conditions.

The dominant wind climate of southern Vancouver Island is characterized by a seasonal reversal between winter and summer winds. Eastern winds are the dominant wind direction from

September through March and can rise to gale, storm, or hurricane force [1]. In the summer months, winds shift to a lighter northwesterly and westerly winds. While winds can travel through the Saanich Inlet, they are much stronger in the open water portions of the Satellite Channel, with gale force winds of 34 to 46 knots periodically forecasted in the Strait of Georgia region [2]. For design purposes, the vessel should be capable of safe operation in winds up to 28-33 knots, with gusts to 41-47 knots.

This route traverses two different wave environments, transiting with the wind changes between the Saanich Inlet and the Satellite Channel. The wave design height will be similar to BC Ferries vessels in the Salish class, as these vessels operate year-round in areas that are very similar to this, as well as sometimes more exposed.

The area lies in a tidal regime very similar to that of Victoria, due to its geographic proximity. The average tidal behaviour in the Victoria region is between 2.7 m and 0.2 m at high and low tides, respectively [3]. As a result, the vessel will be designed to accommodate operation in these conditions. No significant sea ice or icing hazard is anticipated in this environment.

1.5 Requirements

The vessel shall operate a minimum of 8 round-trip sailings per day, with increased frequency during peak commuter periods (06:00–09:00 and 15:00–19:00). Terminal turnaround time, including unloading, loading, and securing of vehicles, shall not exceed 20 minutes at each berth. The vessel shall be a double-ended ferry to eliminate the need for turning manoeuvres in the confined approaches to both terminals, minimizing turnaround time and port infrastructure requirements. The design service life of the vessel shall be a minimum of 30 years.

The vessel shall be designed to carry a minimum of 100 automobile equivalents (AEQ) on the vehicle deck, with the ability to accommodate a mix of standard passenger vehicles, light trucks, and commercial vehicles up to 20 m in length. Passenger capacity shall be a maximum of 450 persons, inclusive of crew. Passenger accommodation shall include an enclosed cabin with seating for at least 300 persons, with the remainder accommodated on open or covered outdoor deck areas. The vessel shall include accessible spaces compliant with Canadian accessibility standards, including elevator access between the vehicle and passenger decks. Bicycle storage for a maximum of 20 bicycles shall be provided separate from the vehicle deck.

1.6 Applicable Regulations

The functional requirements for the design of the passenger vessel are established in accordance with the ABS Rules for Building and Classing Marine Vessels, Part 5C, Chapter 7 [4]. Table 1 summarizes the applicable functional requirements as defined in Section 1.1.2 of [4], which serve as the foundational design criteria for the proposed ferry.

Table 1: Regulatory Requirements Related to Stability and Structure

Functional Requirement No.	Functional Requirements
Stability (STAB)	
STAB-FR1	Side scuttles, windows, and skylights, including the glass, deadlights and storm covers, if fitted, are to be of

	substantial strength to provide watertightness/weathertightness of the external boundary.
STAB-FR2	The position and size of the side scuttles and windows are to minimize the risk of water ingress.
STAB-FR3	The vessel is to be equipped with closing appliances of adequate strength for hull openings.
STAB-FR4	Means of closure are to be provided for openings in locations likely to have water on deck in heavy weather.
STAB-FR5	Suitable tightness of bulkheads is to be provided.
STAB-FR6	Where penetrations are carried through watertight bulkheads and internal decks, arrangements are to be made to maintain the watertight integrity of the bulkheads.
STAB-FR7	Enable the openings/penetrations through watertight boundaries to be closed rapidly.
STAB-FR8	Suitable tightness of tanks is to be provided.
STAB-FR9	Provide adequate means to drain tanks.
STAB-FR10	Provide adequate free ventilation to maintain tank pressure within the designed limits including loading/unloading.
Structural (STRU)	
STRU-FR1	Side scuttles, windows, and skylights, including the glass, deadlights and storm covers, if fitted, are to be of substantial strength to provide watertightness/weathertightness of the external boundary.
STRU-FR2	Adequate longitudinal strength is to be provided to resist bending moments and shear forces associated with vessel loading and wave loads anticipated during the design life.
STRU-FR3	Scantlings are to have sufficient strength to resist failure and excessive deformation associated with buckling and yielding when subjected to the loads anticipated throughout the service life, such as external sea loads, cargo loads, ballast water and any other liquid loads, and accidental overfill during loading.
STRU-FR4	Strength members subject to compressive or shear stresses due to longitudinal bending are to have adequate buckling strength.
STRU-FR5	Stresses in way of large deck openings are to be minimized.
STRU-FR6	The workmanship is to avoid discontinuities, misalignment and other harmful details that may lead to failure in service.

STRU-FR7	Proportions of structural members are to be sufficient to prevent buckling or excessive deflection of structural members.

2. Design Concept

The following sections will outline the proposed design of the vessel outlining key parameters such as hull dimensions and coefficients, area and volume estimates, conceptual sketches, payload, and critical systems onboard the vessel.

2.1 Principal Hull Dimensions & Hull Form Coefficients

The Salish class of BC Ferries is the closest suitable comparison for the proposed Brentwood Bay-Cowichan Bay ferry. These current vessels each measure around 107 m in length, a breadth of 23.5 m, and a maximum displacement of 4227t [5]. Furthermore, they have a carrying capacity of roughly 140 vehicles and up to 600 persons. While the proposed vessel serves a route of approximately 18 km, shorter than current classes, and carries fewer vehicles and fewer passengers, this reference class is still one of the closer suitable comparisons but will be scaled down based off published dimensions of the BC Ferries vessel Salish Eagle.

Dimensional terms are defined per ISO 7462 [6]. Definitions are found in the tables below. The proposed principal vessel dimensions are derived by scaling the Salish Eagle to match the reduced vehicle and passenger capacities, and by erring on the side of caution that there may be shallow water terminal approaches at Brentwood Bay and Cowichan Bay. Furthermore, a double-ended hull is mandated by the operational requirements.

Table 2: Principal Hull Dimensions

Parameter	ISO 7462 Definition	Salish Eagle Value	Proposed Value
Length (L)	The principal longitudinal dimension of a ship.	107 [m] [5]	88.0 [m]
Breadth (B)	A dimension expressing the width of a ship in a transverse horizontal direction.	23.5 [m] [7]	20.0 [m]
Depth (D)	The principal vertical dimension of a ship's hull.	6.60 [m] [8]	6.5 [m]

Table 3: Estimated Hydrostatic Conditions

Parameter	ISO 7462 Definition	Salish Eagle Value	Proposed Value
Displacement Mass	The mass of water displaced by a ship when floating statically at a specified draught and trim.	4227 [t]	3641 [t]
Draught Amidships (Draft)	Extreme draught measured at the after perpendicular	4.65 [m] [9]	4.1 [m]

Given the density of saltwater (1.025 t/m^3), the displacement volume can be determined.

$$\nabla = \frac{W}{\rho} = \frac{3641}{1.025} = 3552\ m^3$$

The Block Coefficient calculation can be seen below:

$$C_b = \frac{\nabla}{L * B * T} = \frac{3552}{88 * 20 * 4.1} = 0.49$$

The midship coefficient is taken as $C_m = 0.82$ from the hydrostatics report.

$$C_p = \frac{C_b}{C_m} = \frac{0.49}{0.82} = 0.60$$

The resulting block coefficient of 0.49 and prismatic coefficient of 0.60 are slightly finer than the typical ranges of 0.55–0.60 (block) and 0.62–0.68 (prismatic) cited for ferry vessels [10]. This reflects the fine-ended form of the double-ended parent hull; the values remain representative of a slender commuter ferry and are considered acceptable at this preliminary design stage.

2.2 Area & Volume Summary

The following area and volume estimates were developed following the feasibility study methodology described by Section 5.3.2.5 in [10], in which the total required internal volume is estimated as a gross figure reflecting payload volume plus the volume required for crew living, propulsion machinery, tankage, stores, and other miscellaneous spaces. The Salish Eagle serves as the parent vessel for scaling these estimates. Compartment allocations are preliminary and will be refined as the design progresses through subsequent iterations of the design spiral [11].

The vessel is arranged across five principal deck levels plus wheelhouses, closely following the Salish Eagle's vertical arrangement. Deck 1 comprises the double-bottom tank spaces and machinery below the lower vehicle deck. Deck 2 is the lower vehicle deck, accommodating standard passenger vehicles with a restricted overhead clearance of 2.3 m. Deck 3 is the main vehicle deck at the freeboard deck level, providing full-height drive-through vehicle stowage for both standard and commercial vehicles. Deck 4 is the enclosed passenger deck, housing all interior passenger accommodation, amenities, and crew spaces. Deck 5 is an open sun deck providing outdoor seating and life raft stations. Wheelhouses are located at the forward and aft ends above Deck 5 to permit double-ended operation.

The Salish Eagle employs the same two-vehicle-deck and two-passenger-deck arrangement to achieve its 138-vehicle and 600-passenger capacity. The proposed vessel retains the two vehicle decks but consolidates passenger accommodation onto a single enclosed deck with an open sun deck above, consistent with the reduced 450-passenger requirement on a shorter commuter route.

The hull depth has been set at D = 6.5 m to accommodate the two-vehicle-deck arrangement within the hull envelope. The lower vehicle deck is located above the machinery space at approximately 2.5 m above baseline, and the main vehicle deck is at the freeboard deck level at 6.5 m above baseline. The main vehicle deck usable envelope is approximately 74 m × 17 m,

deducting 7 m at each end for bow and stern ramp structure and 1.5 m per side for hull structure and safety walkways. The lower vehicle deck usable envelope is reduced to approximately 60 m × 15 m, reflecting the narrowing of the hull form below the main deck and the space occupied by internal vehicle access ramps. The Deck 4 superstructure footprint is approximately 55 m × 19 m, and the Deck 5 sun deck footprint is approximately 50 m × 17 m. See the area and volume estimates below in Table 4.

Table 4: Area and Volume Summary

Compartment	Deck	Area (m²)	Height (m)	Volume (m³)	Purpose
Double-bottom tanks	1	1,100	1.4	1,540	Fuel oil, freshwater, and ballast water storage
Machinery space	1	280	2.5	700	Main engines, generators, switchboards, and auxiliary machinery
Steering gear rooms (×2)	1	120	2.5	300	Steering gear and bow/stern thruster machinery at each end
Lower vehicle deck	2	900	2.3	2070	Vehicle stowage for standard passenger vehicles only (maximum 2.1 m vehicle height); approximately 50 AEQ
Internal vehicle ramps (×2)	2-3	80			Ramps connecting lower and main vehicle decks at each end
Main vehicle deck	3	1,258	4.5	5,661	Vehicle stowage and loading for standard and commercial vehicles; approximately 72 AEQ with drive-through fore-and-aft lanes
Bicycle storage	3	30	2.4	72	Secured storage for maximum 20 bicycles, separated from vehicle lanes

Enclosed passenger cabin	4	500	2.4	1,200	Seated accommodation for 300 passengers with accessible seating areas
Food service	4	60	2.4	144	Passenger food and beverage service
Washrooms	4	50	2.4	120	Passenger and accessible washroom facilities
Crew quarters	4	80	2.4	192	Crew rest area, lockers, and mess for up to 12 crew
Covered walkways and exterior circulation	4	150	2.4	360	Exterior covered passages around the perimeter of Deck 4
Open sun deck	5	400	-	-	Open-air seating and viewing areas for up to 150 passengers
Life raft stations	5	60	-	-	Life raft stowage and deployment areas
Stairwells, elevator, and passages	2–5	120	-	480	Vertical and horizontal circulation including elevator between vehicle and passenger decks
Forward wheelhouse	Above 5	48	2.6	125	Navigation, helm, and communications for forward operation
Aft wheelhouse	Above 5	48	2.6	125	Navigation, helm, and communications for aft operation
Totals		**5,284**		**13,089**	

Open and exterior areas are reported by area only as they contribute no enclosed volume. The lower vehicle deck clearance of 2.3 m restricts this deck to standard passenger vehicles with a maximum height of 2.1 m; all commercial vehicles and light trucks exceeding this height are directed to the main vehicle deck. The main vehicle deck height of 4.5 m is set by the commercial vehicle clearance requirement established in Phase 0.

2.3 Configuration Sketches

In accordance with the dimensions outlined in Sections 0 and 0, the following figures outline the critical dimensions of the vessel. Figure 2 and Figure 3 display the critical dimensions of the entire vessel including the draft, breadth, depth, design waterline (DWL), freeboard, and baseline.

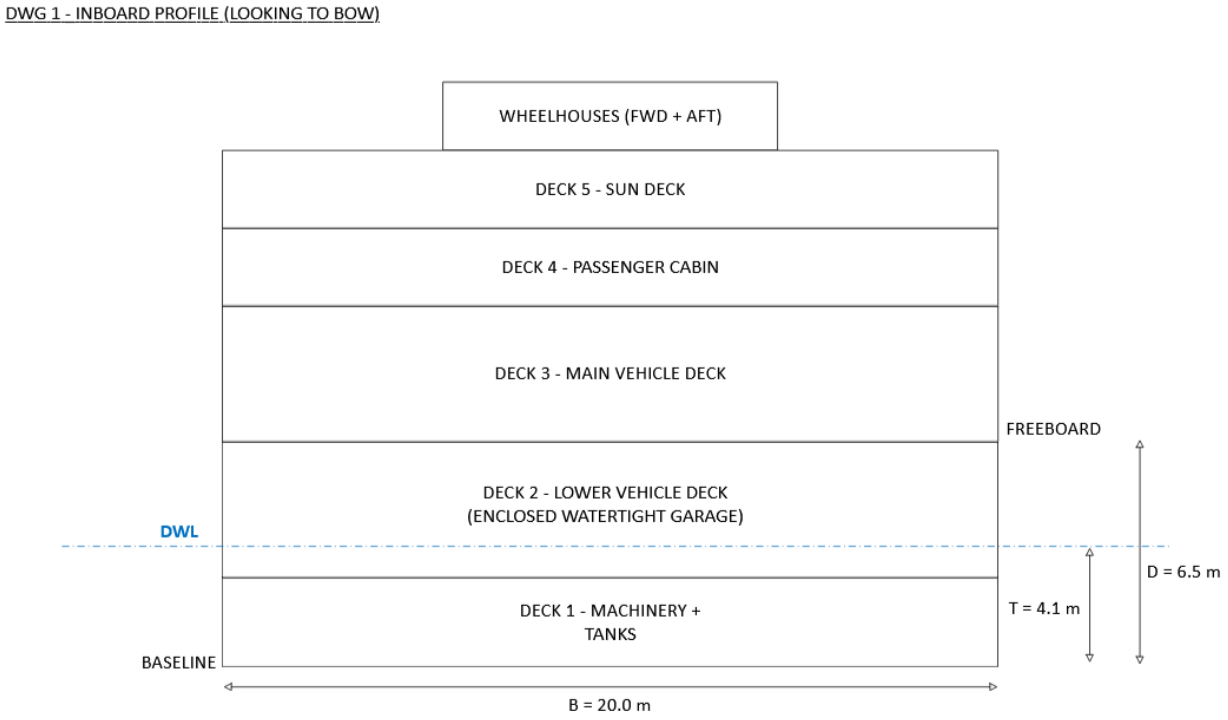


Figure 2: Ship Profile (Looking to Bow)

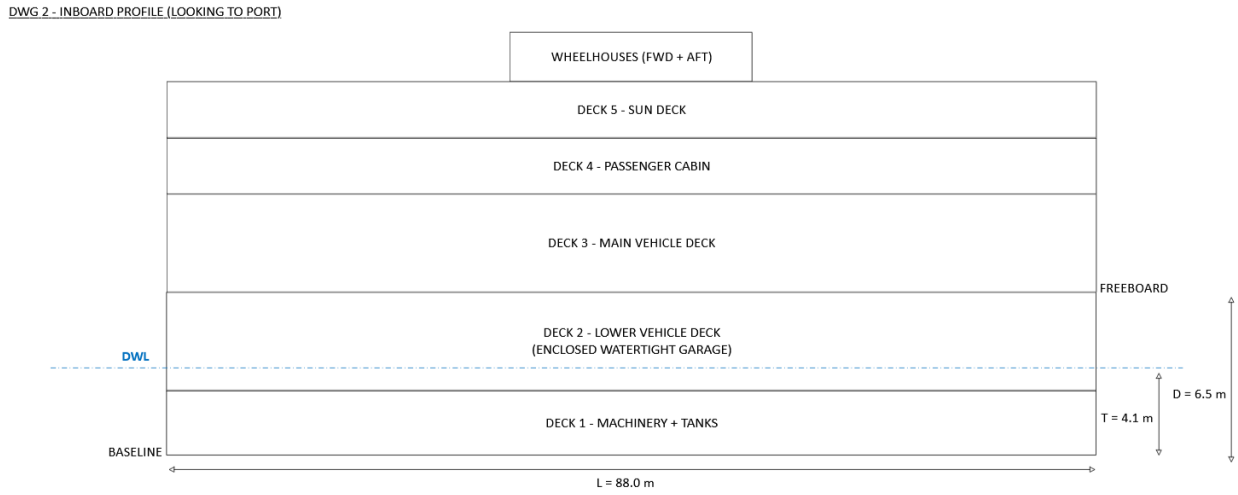


Figure 3: Ship Profile (Looking to Port)

Figure 4 and Figure 5 portray the proposed vehicle deck plans. As previously described, Deck 2 is limited to passenger vehicles only, while Deck 3 contains area for both passenger and commercial vehicles. It is important to note that since the base of Deck 2 is above the water line, Deck 2 will be designed as a watertight garage. This will be discussed more in-depth in Section 2.5. The useable areas of each deck correspond to the dimensions outlined in Section 2.2.

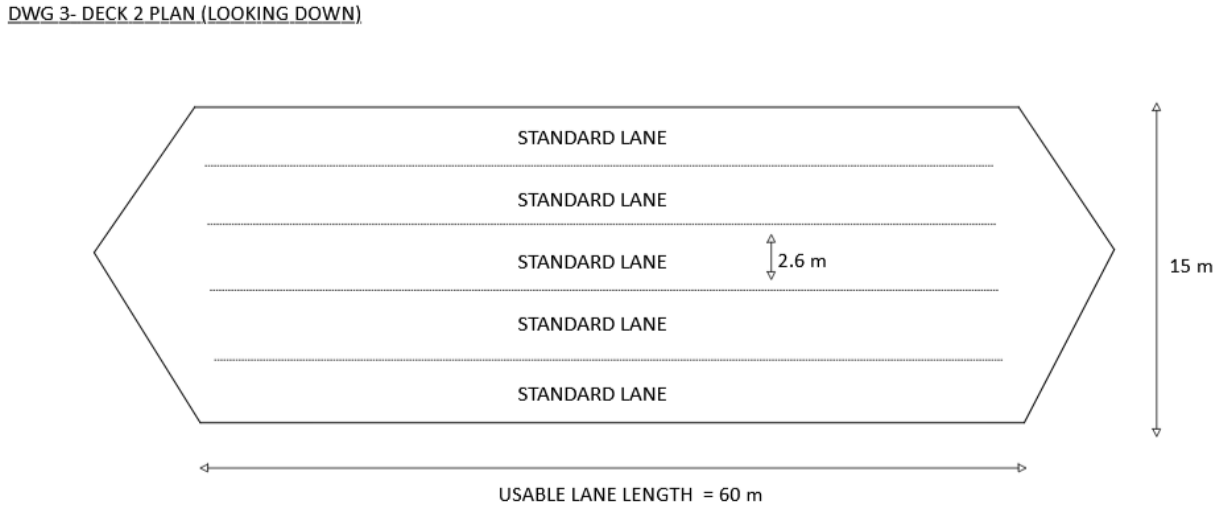


Figure 4: Deck 2 Plan (Looking Down)

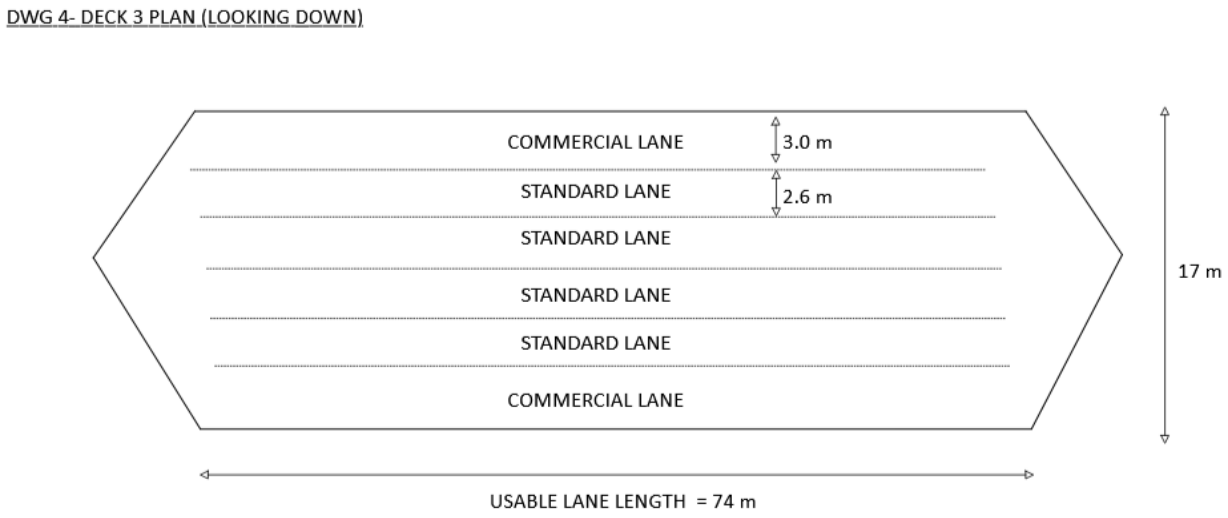


Figure 5: Deck 3 Plan (Looking Down)

2.4 Payload Definition

The vessel is designed to carry a maximum of 450 persons inclusive of crew, as established in the previously outlined requirements. For payload weight calculations, a standard mass of 75 kg per person will be applied. A luggage allowance of 15 kg per person will be assumed, reflecting

the mixed commuter and recreational use of the route, where some passengers carry minimal items and others travel with larger loads.

Table 5: Weight Tonnage due to Passengers

Item	Quantity	Unit Mass [kg]	Total [kg]
Passengers (incl. crew)	450	75	33,750
Luggage	450	15	6,750
Totals			**40,500**

The vehicle deck is designed for a minimum of 100 AEQ, accommodating a mix of standard passenger vehicles, light trucks, and commercial vehicles up to 20 m in length. A representative loading mix is assumed below to reflect typical traffic on a commuter and commercial route. The AEQ model assumes one standard car equals 1 AEQ, with larger vehicles occupying multiple AEQ of total deck space.

Table 6: Weight Tonnage due to Vehicles

Vehicle Type	Quantity	AEQ Each	AEQ Total	Unit Mass [kg]	Total Mass [kg]
Standard passenger cars	70	1	70	1,800	126,000
Light trucks/vans	15	1.5	22.5	3,000	45,000
Commercial vehicles	5	4	20	25,000	125,000
Totals	**90**		**112.5**		**296,000**

The assumed vehicle mix yields a 112.5 AEQ satisfying the 100 AEQ minimum with some margin. Vehicle masses are representative averages [12]; the commercial vehicle mass reflects a partially loaded goods vehicle rather than a fully loaded semi-trailer at maximum gross weight.

In addition, an average weight of 15 kg has been assumed for a bicycle. With storage for a maximum of 20 bicycles, the total weight added to the payload due to bicycles is assumed to be approximately 300 kg.

Table 7: Design Payload

Component	Mass [t]
Passengers	33.75
Luggage	6.75
Vehicles	296
Bicycles	0.3

Total Payload	336.8

This total represents the minimum design payload in which the vessel must be able to accommodate.

2.5 Critical Space Requirements

The vessel features two vehicle decks. The main vehicle deck (Deck 3) is located at the freeboard deck level and accommodates all vehicle types including commercial vehicles. The lower vehicle deck (Deck 2) is located within the hull below Deck 3 and is restricted to standard passenger vehicles only.

On the main vehicle deck (Deck 3), lane width is set at a minimum of 2.6 m per standard lane to allow door clearance and safe boarding, with commercial vehicle lanes widened to 3.0 m. The standard vehicle footprint is taken as 6.1 × 2.6 m, matching the Salish-class reference vessel; this is consistent with the BC Ferries definition of a standard vehicle as one under 7 ft (2.13 m) high and less than 20 ft (6.10 m) in length [13], with larger vehicles assigned to the commercial lanes. Deck 3 clearance is set at a minimum of 4.5 m to accommodate the commercial vehicles specified. Vehicle lanes run continuously fore-to-aft with drive-through loading at both ends, eliminating the need for turning areas. On the lower vehicle deck (Deck 2), lane width is maintained at 2.6 m. Overhead clearance is set at 2.3 m, restricting this deck to standard passenger vehicles with a maximum height of 2.1 m. As this deck lies partly below the design waterline, it is arranged as an enclosed, watertight garage with weathertight closures at the internal ramp connections, maintaining the watertight integrity of the freeboard deck above. Commercial vehicles, light trucks, and any vehicle exceeding 2.1 m in height are directed to Deck 3. Access to Deck 2 is provided by internal vehicle ramps connecting to Deck 3 at the forward and aft ends. The reduced usable envelope of Deck 2 (approximately 60 m × 15 m) reflects the narrowing of the hull form below the main deck and the space occupied by the access ramps.

Passenger cabin deck headroom is set at a minimum of 8 ft (2.4 m). Bicycle storage for a maximum of 20 bicycles is provided in a dedicated area separate from the vehicle deck, with securing points to restrain bicycles during travel.

2.6 Mission-Critical Systems & Features

This section identifies the systems and features essential to the safe and reliable operation of the vessel. Each system is described in terms of its purpose, general location on the vessel, and the regulation that governs it. The following sections (in conjunction with the regulatory requirements outlined in Section 1.6) will serve as the basis for the design criteria of the vessel.

2.6.1 Fire Safety

Fire is one of the most significant hazards to life aboard a passenger vessel. The open vehicle decks carry vehicles with fuel tanks and electrical systems in a large, undivided space, while the passenger cabin holds up to 450 persons who must be protected and, if necessary, evacuated. The

enclosed lower vehicle deck (Deck 2) increases this risk, as a fire in a below-deck garage is harder to contain. The functional requirements in Table 8 establish the fire-safety design criteria, as outlined in the governing ABS Rules document [4].

Table 8: Fire Safety Functional Requirements

Functional Requirement No.	Functional Requirements
Structure	
STAB-FR1	Materials used in the ships' structure shall ensure that the structural integrity is not degraded due to fire. (SOLAS II-2/11.1).
STAB-FR2	Separation of accommodation spaces from the remainder of the ship by thermal and structural boundaries (SOLAS II-2/2.2.1.2)
Materials	
MAT-FR1	The ignitability of combustible materials shall be restricted. (SOLAS II-2/4.1.3)
MAT-FR2	The use of combustible materials shall be restricted. (SOLAS II-2/2.2.1.3, SOLAS II-2/5.1.3)
MAT-FR3	The quantity of smoke and toxic products released from combustible materials, including surface finishes, during fire shall be limited. (SOLAS II-2/6.1)
Fire Safety (FIR)	
FIR-FR1	Means for controlling smoke in atriums, control stations, machinery spaces and concealed spaces shall be provided. (SOLAS II-2/8.1)
FIR-FR2	The ship shall be subdivided by thermal and structural boundaries. (SOLAS II-2/2.2.1.1, SOLAS II-2/9.1.1)
FIR-FR3	Thermal insulation of boundaries shall have due regard to the fire risk of the space and adjacent spaces. (SOLAS II-2/9.1.2)
FIR-FR4	The fire integrity of the divisions shall be maintained at openings and penetrations. (SOLAS II-2/9.1.3)
FIR-FR5	Protection of means of escape and access for fire fighting to enable sufficient time for escape and evacuation. (SOLAS II-2/2.2.1.6)
FIR-FR6	Containment of any fire in the space of origin. (SOLAS II-2/2.2.1.5, SOLAS II-2/9)
Escape, Evacuation, Rescue (EER)	
EER-FR1	Safe escape routes shall be provided. (SOLAS II-2/13.1.1)
EER-FR2	Escape routes shall be maintained in a safe condition, clear of obstacles. (SOLAS II-2/13.1.2)

EER-FR3	Additional aids for escape shall be provided as necessary to ensure accessibility, clear marking, and adequate design for emergency situations. (SOLAS II-2/13.1.3)

These functional requirements establish the fire safety design criteria for the vessel. The structure and materials requirements ensure that the hull retain integrity and limit spread of fire. The fire safety requirements address detection and containment, while the escape, evacuation, and rescue requirements ensure that the full onboard capacity can be safely evacuated.

2.6.2 Piping & Bilge Systems

The piping and bilge systems govern the vessel's ability to survive flooding and to operate without polluting the environment. Flooding survivability is especially critical for this vessel because the lower vehicle deck lies partly below the design waterline; any water ingress into this area must be detected and removed before it can cause stability issues. The bilge system provides the ability to drain liquids and to control flooding, while the handling requirements prevent discharge into the Saanich Inlet. The functional requirements most relevant to flooding and environmental protection are shown in Table 9, where the complete set can be found in the ABS Rules document [4].

Table 9: Piping & Bilge Systems Functional Requirements

Functional Requirement No.	Functional Requirements
Stability (STAB)	
STAB-FR2	Avoid the possibility of cross-flooding between the vessel and the sea.
STAB-FR3	Provide means to dispose of or clear accumulated liquids in spaces within the vessel due to condensation, leakage, washing, firefighting.
STAB-FR5	Provide means to enhance the bilge system availability in case of loss of compartment or bilge equipment, considering safety and number of passengers onboard.
STAB-FR9	Valves required to control external flooding, and their controls are to be readily accessible and suitably arranged to enable safe operation by the crew.
STAB-FR10	Provide means to detect flooding for all watertight spaces considering the number of passengers onboard.
Protection of Environment (ENV)	
ENV-FR1	Provide means to discharge sludge and handle oily bilge water to prevent polluting the sea.
Safety of Personnel (SAFE)	
SAFE-FR1	Provide means to control flooding in any watertight compartment (other than spaces appropriated for fresh water, ballast, fuel oil or liquid cargo) resulting from damage to piping systems.

2.6.3 Electrical Systems

The electrical system supplies power to the vessel's propulsion, navigation equipment, lighting, passenger systems, and life-safety systems. A loss of power on a vessel carrying 450 persons in the confined waters of Saanich Inlet could disable steering, navigation, and evacuation lighting simultaneously. The design therefore requires a main power source supported by an independent emergency source that remains available even if the main machinery source is compromised. The functional requirements in Table 10 establish the criteria. For the full list of requirements, refer to the ABS Rules document [4].

Table 10: Electrical Systems Functional Requirements

Functional Requirement No.	Functional Requirements
Power Generation & Distribution (POW)	
POW-FR1	Provide sufficient power capacity and redundant quantity to achieve continuity of power for essential services and comfortable conditions of habitability.
POW-FR2	Power generation and distribution equipment to be designed with redundancy to prevent loss of essential/emergency services upon a single failure.
POW-FR4	Provide a readily available power source independent of the main source of power to support emergency services for the applicable duration.
POW-FR5	Provide arrangement and location provisions for emergency power such that a casualty of the space containing the main source of power will not affect the emergency power.
POW-FR6	Provide transitional power to supply critical safety systems that are not to be interrupted upon loss of either the main or emergency source of power.
Escape, Evacuation, Rescue (EER)	
EER-FR1	Provide manual and effective means at readily accessible locations to alert/notify of fire and its position upon discovery.
EER-FR2	Enable communication to be broadcast to persons onboard to notify them of the emergency and the actions to undertake.
EER-FR3	Arrangements are to be provided to enable passengers to readily identify all escape exits and escape routes.
EER-FR4	Provide additional lighting at locations that are essential and can assist persons on board in any escape, evacuation and rescue activity.

Together, these functional requirements define the design criteria for the vessel's mission-critical systems: fire protection of the vessel, flooding survivability, and continuity of electrical power to

safety-related systems. Each will be developed into specific equipment selections and arrangements in the following design phases.

3. Preliminary Weight Estimates and Powering

Upon completion of the formation and scaling process in DELFTship, new hydrostatic properties were calculated for the Parent 32 hull, which was used as the basis of the hull formation for this ship. This final value was calculated to be 3640.952 tonnes, which represents the vessel's total displacement at a draft of 4.1 m. However, this is not a representation of the lightweight or deadweight values. Instead, this is a calculation performed by DELFTship which includes taking the final hull formation, slicing it into stations along its length, and then integrating these sectional areas along the length of the hull, up to the 4.1 m waterline. The result is the volume of water which will be displaced by this draft value, given as the moulded volume of 3552.15 m^3. Multiplying this by the density of seawater gives the calculated displacement value. As this number is purely a geometric result, it reflects only the shape of the hull below the specified waterline. As this value represents the weight the vessel must have to float at the draft, it is usable for deadweight and lightship weight estimates. However, estimates for these splits are required, and are developed in the remainder of this section.

Biran notes that in the first stages of ship design, lightship masses are either derived from empirical equations or based on the masses of a given parent ship [14]. This report adopts the Salish Eagle as the parent ship given its close comparability to the proposed vessel. It should be noted that this is distinct from the "Parent 32" hull form used in DELFTship to generate the preliminary vessel geometry and curves. The parent ship referenced here is a real vessel.

The Salish Eagle has a published displacement of 4227 tonnes and a maximum deadweight of 864 tonnes [7]. As a result, the estimated lightweight value of this ship can be found by calculating:

$$Lightship = \Delta - Deadweight$$

$$3363t = 4227t - 864t$$

This means the Salish Eagle's lightship fraction is approximately 79.6%. Applying this to the vessel's DELFTship displacement of 3640.95t:

$$Lightship = 0.796 * 3640.95 = 2898t$$

$$Deadweight = 0.796 * 864t = 743t$$

As a result, estimated lightship and deadweight values are summarized below:

Table 11: Lightship & Deadweight Estimations

Property	Value
Estimated Lightship Value	2898t
Estimated Deadweight Value	743t

For propulsion, the vessel is proposed to use a dual fuel LNG/diesel fuel system. This is due to the fact that surrounding local ferries already utilize LNG and operate on the same general Salish Sea/Gulf Islands routes as the proposed service. LNG is well suited to this application as BC has an abundant local supply from the province's natural gas industry, along with infrastructure already available from the neighboring fleets. Furthermore, it provides air quality and carbon emission benefits relative to conventional diesel fuel [15].

4. Hull Form

The hull form presented in this section was developed in DELFTship using a parent-hull scaling approach. Parent 32, a single-ended ferryboat form from the program database, was selected as the closest available match to the proposed vessel's mission and configuration for its representative ferry hull geometry. It was scaled to the principal dimensions established in Section 2.1: a length of 88.0 m, a breadth of 20.0 m, and a design draught of 4.1 m. Because the parent hull did not satisfy the double-ended requirement, the form was modified to be symmetric about the midship plane. The hull was then iterated against the design coefficients outlined in Section 2.1 until the results converged in accordance with the design requirements. Hydrostatic properties were verified against the design condition, yielding a displacement of 3641 tonnes at the design draught.

From this model, the following hull-form deliverables were generated and will be discussed throughout this section. Together these define the geometry of the underwater and above-water hull and provide the basis for the stability, powering, and structural analyses.

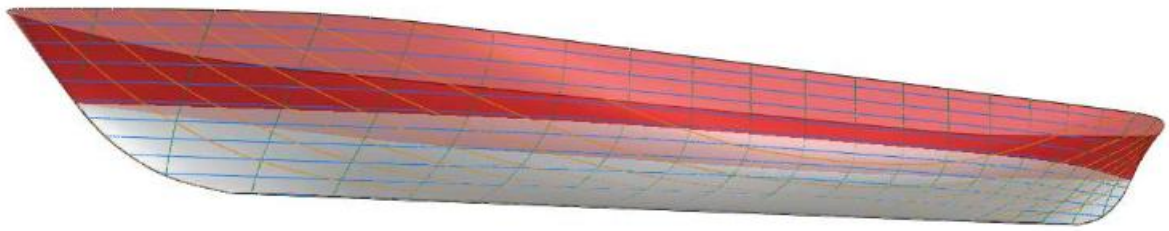


Figure 6: Hull Form

4.1 Line Plans

The following section outlines the lines plans for the proposed vessel which represents the complex 3D shape of the ship's hull. The figures will outline the shear profile, body, and half-breadth plans. A complete lines plan document including all views can be found in Appendix B. One important note; since the fore and aft ends of the vessel are mirrored about the midship, the presented body plan (Figure 8) is representative of both ends of the vessel. As the following images may be difficult to decipher under scale, a separate document has been attached.

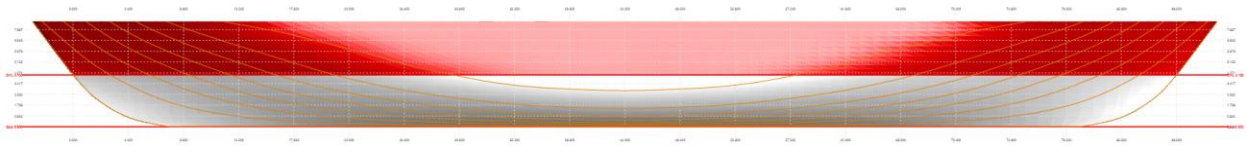


Figure 7: Shear Profile Plan

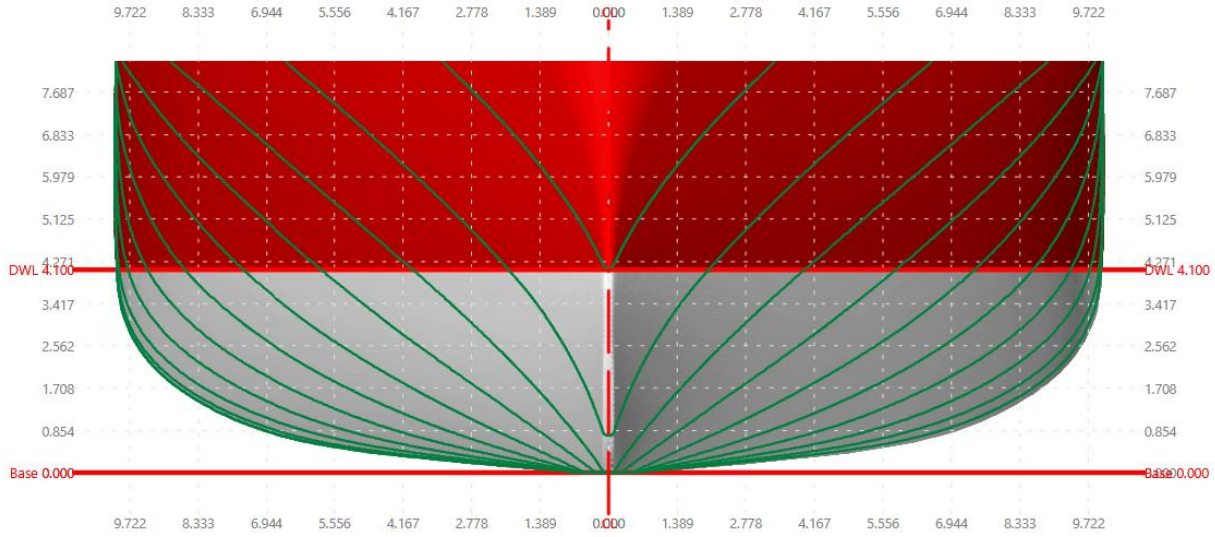


Figure 8: Body Plan

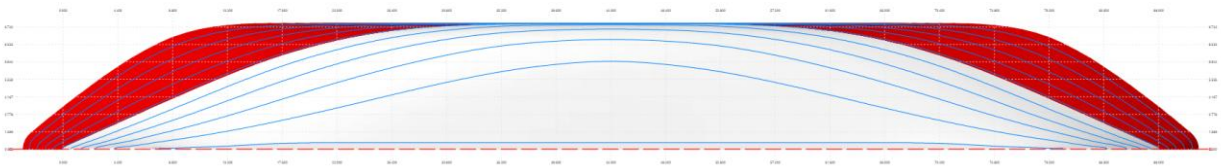


Figure 9: Half-Breadth Plan

4.2 Hydrostatic Curves

The hydrostatic curves, or curves of form, describe how the vessel's hydrostatic properties vary with draught. They were generated in DELFTship from the scaled parent hull over a range of draughts from 0 m to 5.0 m, with the vessel on an even keel. Each curve is plotted against draught on the vertical axis, allowing the corresponding value of each parameter to be read off at any waterline, including the design draught of 4.1 m.

The curves presented include the displacement, longitudinal and vertical centres of buoyancy (LCB, VCB), longitudinal centre of flotation (LCF), transverse and longitudinal metacentric heights, waterplane area, tonnes per centimetre immersion (TPC), and the moment to change trim. These quantities form the basis for the vessel's stability, trim, and loading assessments in subsequent design phases. At the design draught of 4.1 m, the curves are consistent with the principal hydrostatic condition established in Section 2.1, giving a displacement of 3641 tonnes. See the curves in Figure 10 below.

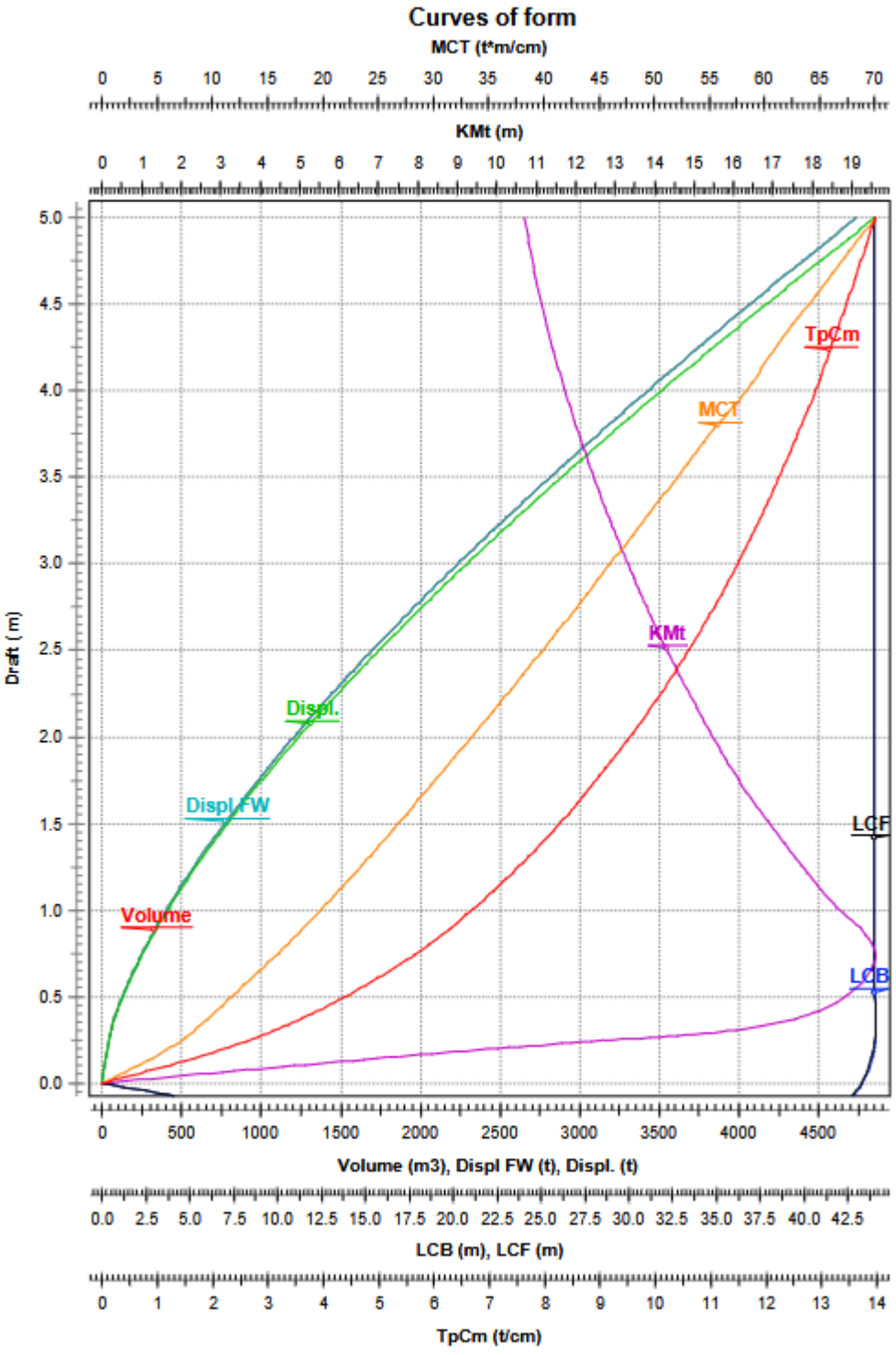


Figure 10: Hydrostatic Curves

4.3 Bonjean Curves

The Bonjean curves describe the immersed cross-sectional area of the hull at each station as a function of waterline height. They were generated in DELFTship from the scaled parent hull for all stations along the length; the complete set of tabulated values is provided in Appendix A. In accordance with the plotting requirement for this phase, the midship section (Station 44.0) is presented graphically in Figure 11 as the representative Bonjean curve.

The curve is plotted with immersed sectional area on the vertical axis and height above the baseline on the horizontal axis, allowing the submerged area at the midship section to be read at any draught. At the design waterline of 4.1 m, the midship section has an immersed area of 67.25 m², consistent with the midship section area reported in the vessel's hydrostatic condition. Because the vessel is a symmetric double-ender, the Bonjean curves for stations equidistant forward and aft of midships are identical, as reflected in the tabulated values found in Appendix A.

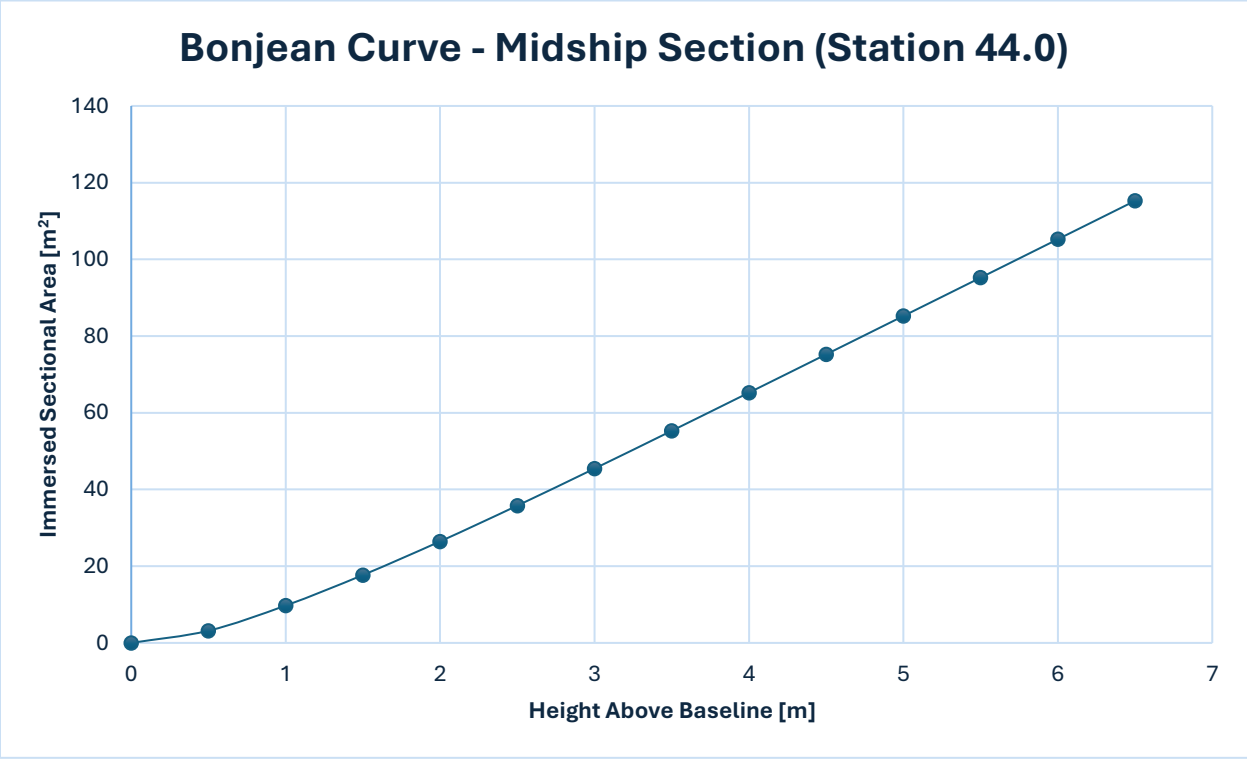


Figure 11: Bonjean Curve for Station 44 (Midship)

4.4 Lightship & Deadweight Estimates

As mentioned in the previous section of this document, Lightship calculations were performed by scaling the parent hall Salish Eagle to the appropriate size. This scaling approach was used as an early approximation before the hull form had been fully modelled. With the hull surface now modelled directly in DELFTship, a more accurate lightship weight was calculated and summarized in the design hydrostatics report. This resulted in a total lightship weight of 2666 t.

For the deadweight calculation, passengers, luggage, vehicles and bicycle estimations remained consistent with the previous sections in this report but are summarized for convenience below. However, propulsion systems were specified to get an accurate weight estimate in DELFTship. A dual fuel engine uses as it is would be capable of retaining the well-established LNG bunkering infrastructure serving the Salish Sea/Gulf Islands ferry fleet.

Following the same scaling approach adopted in previous sections, installed propulsion power was scaled. The Salish class power is reported as 4438 kW [16]. Using the same displacement, the power can be calculated:

$$\frac{P_{Proposed}}{\Delta_{Proposed}} = \frac{P_{Salish}}{\Delta_{Salish}}$$

$$P_{Proposed} = \frac{P_{Salish}}{\Delta_{Salish}} * \Delta_{Proposed} = 4438 * \frac{3641}{4227} = 3823\ kW$$

As this does not separately account for the smaller difference in design speed, which is 14.5 knots instead of the 15.5 knots of the Salish class ferries, this would expect to be an overestimation, of power, while still being reasonable for this design stage.

Three Wärtsilä 20DF-platform dual fuel medium speed engines are proposed as generating sets. This engine is already in service in the Salish class which gives crew familiarity for maintenance. These units are rated at 1,200 kW each, providing a total of 3,600 kW of target capacity.

To perform the weight estimation, manufacturer data for the generating set lists a dry weight of 17 t [17]. The total propulsion machinery weight was then built by component. Lubricating oil changes are included in the rows below, but fuel is excluded as it is accounted for separately.

Table 12: Propulsion System Weight Estimate

Component	Weight (t)
3x Wärtsilä 6L20DF, including alternator and base plate	50 [17]
Bow thruster unit	30 [18]
Stern thruster unit	30
Fueling supply system (piping, valves, etc.)	30
Exhaust Gas Systems	18
Coolant Systems	12
Machinery Mounting Components and Seatings	20
Miscellaneous Spare Weight Allowance	35
Lubricating Oil Systems	10
Engine Electronic System Components	40
Total Weight	**275**

Table 4 allocates 1549 cubic meters of double bottom volume to combined fuel oil, freshwater, and ballast storage. This section derives the individual capacity of each for the deadweight estimate.

Ballast capacity for a short sea ferry is estimated as a percentage of deadweight. A ballast allowance of 20% is used for this calculation. Applying this:

$$W_{Ballast} = 0.20 * 743t = 149.2t$$

Portable water demand is estimated from passenger and crew capacity. A design allowance of 40L/person/day is applied to the 450 persons estimated, with a 1.5 day reserve.

$$W_{Freshwater} = 450 * 40L * 1.5 * 1\frac{t}{m^3} = 27t$$

The fuel estimate can be found from estimating the mass flow rate of fuel, from the estimated engine size. Using the Wartsila 20 engine and an assumed average load of 70% for the 3600kW (giving 2,520kW), with a fuel rate of 0.189 g/kWh [19]:

$$\dot{m}_{fuel} = 2520\ kW * 0.189\frac{g}{kWh} = 0.469\frac{t}{h}$$

The assumed weight for fuel is 115.5t. This was due to a budgeted liquid mass amount of 142.5t during initial design, with 27t used up on freshwater. Checking the amount of time this would supply fuel for gives:

$$t_{buffer} = \frac{115.5t}{0.469\frac{t}{h}} = 246\ hours = 10.3\ days$$

Giving roughly 10 days of full diesel backup, which is a high amount but accounts for multiple days of no diesel availability. They are summarized in the table below:

Table 13: Liquid Mass Breakdown

Tank	Weight (t)
Ballast Water	149.2
Freshwater	27
Fuel	115.5

Finally, the deadweight + lightship estimate is placed below:

Table 14: Deadweight + Lightship Estimate

Weight Summary	tonnes
Lightship (Estimated)	2666
Passengers	33.8
Luggage	6.8
Vehicles	296
Bicycles	0.3

Propulsion	275
Ballast Tanks	149.2
Fuel & Freshwater	142.5
Δ = Lightship + Deadweight	**3,569.6**

4.5 Machinery & Tank Holds

This section outlines the machinery and tank configurations, placement, and dimensions onboard the vessel. The machinery and tanks are organized into four categories:

- Propulsion
- Ro-Ro space (vehicle decks)
- Ballast tanks
- Liquid cargo (fuel and freshwater)

Based on the proposed dual LNG-diesel propulsion system, the machinery space is allocated directly below Deck 2, spanning the full depth of Deck 1 (baseline to 2.5 m). This placement keeps the propulsion system centered near midship, consistent with the machinery space footprint defined in Table 4, and allows direct routing to any fore/aft systems given the vessel's double-ended configuration.

Ro-Ro (roll-on/roll-off) space was determined early in the design process and is represented by Deck 2 and Deck 3. Deck 2, the lower vehicle deck, is restricted to standard passenger vehicles under an enclosed, watertight garage arrangement, while Deck 3, the main vehicle deck, accommodates the full mix of standard and commercial vehicles with drive-through access at both ends. This two-deck arrangement reflects the vertical placement of the vehicle payload weights used in the vessel's loading conditions.

Two separate ballast configurations were considered: double-bottom ballast tanks and fore/aft peak ballast tanks. Given the limited available space along the bottom of the vessel (largely occupied by machinery and liquid cargo) and consistent with standard convention in large vessel design [20], fore/aft peak ballast tanks were selected for this design. The tanks run the full depth of the hull, from the keel to the bulkhead (freeboard) deck. The tanks are symmetrically mirrored about midship, reflecting the vessel's double-ended hull form.

Liquid cargo space was allocated for fuel and freshwater storage. Fuel storage was placed forward of the engine room, while freshwater was placed aft. Fuel tank volume was allocated using the remaining available double-bottom space between the engine room, the fore peak ballast tank, and the underside of Deck 2. Freshwater tank volume was approximated based on the vessel's 450-person design payload, as outlined in the previous section.

The placement of all machinery and tanks can be found in the figure below. The complete tank capacities report can be found in Appendix E.

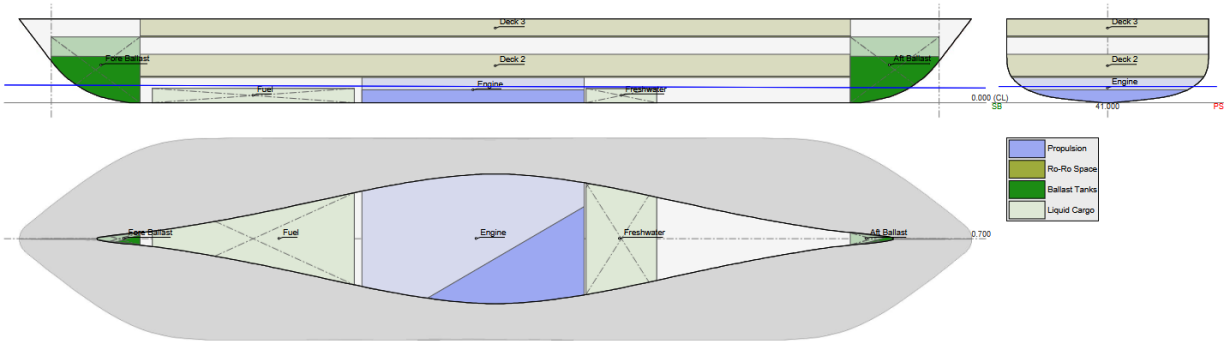


Figure 12: Tank Arrangement

A breakdown of the total tonnage volume was obtained from the tonnage calculation feature in DELFTship. This represents the available space inside each of the allocated machinery and tank volumes.

Input data

Draft	6.500 m
Depth	4.100 m
Number of passengers in cabins with not more than 8 berths	0
Number of other passengers	450

Included spaces

Hull	0.000 m³
	0.000 m³
Total	**0.000 m³**

Cargo spaces

Engine	749.085 m³
Deck 2	2632.856 m³
Aft Ballast	190.481 m³
Fore Ballast	190.273 m³
Freshwater	87.518 m³
Fuel	135.248 m³
Total	**3985.460 m³**

Results of the calculation

Total enclosed volume	0.000 m³
Gross tonnage	**0**
Net tonnage	**1140**

Figure 13: Tonnage Calculation

4.6 Floodable Length & Freeboard

With tank/weight items and a margin line established (set 76 mm below the bulkhead deck), the floodable length of the vessel was calculated in DELFTship. The complete floodable length data tables are provided in Appendix D, while Figure 14 presents the resulting floodable length curve for the vessel.

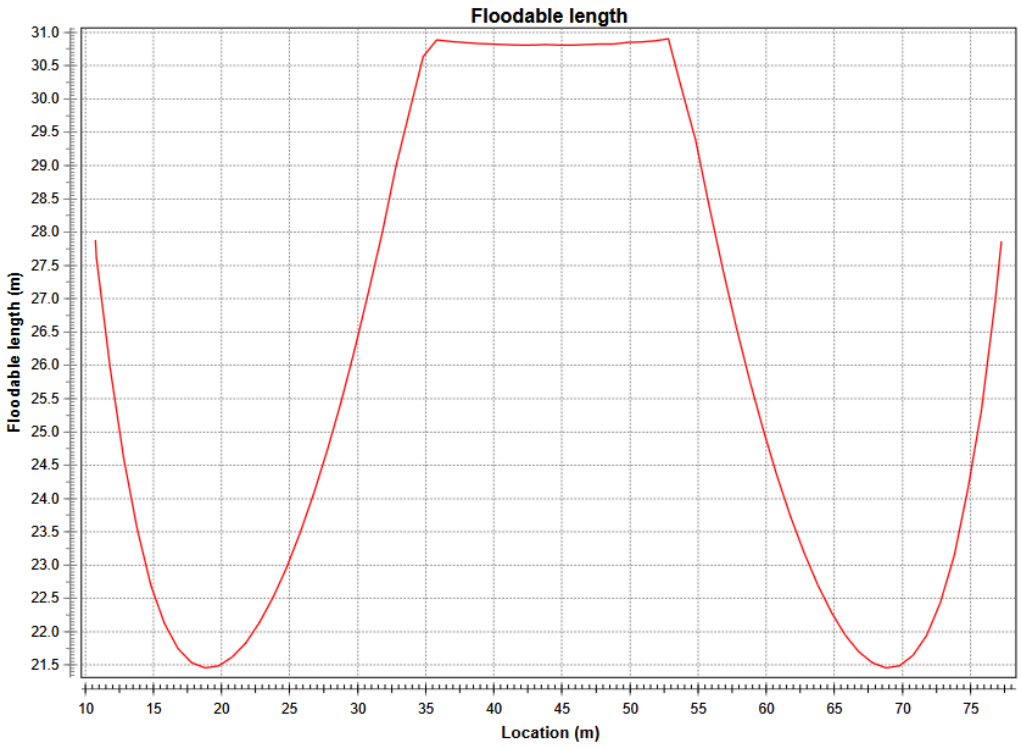


Figure 14: Floodable Length Plot

The floodable length curve for the vessel, displays a symmetric M-shape about midship, consistent with the double-ended hull form. Two local minima of approximately 21.5 m occur near stations 19 m and 69 m, while a plateau of roughly 30.8 m is maintained amidships (stations 35–53 m). The curve decreases further toward the bow and stern, reflecting reduced reserve buoyancy in the narrower end sections. The vessel's fore and aft peak ballast tanks, each approximately 13.2 m long, remain well within the floodable length limit at their location.

With the floodable length deemed acceptable, the freeboard numbers can now be produced. Given a depth of 6.5 m and a design draught of 4.1 m, the resulting Summer (S) freeboard is:

$$F_{summer} = D - T = 6.5\text{ m} - 4.1\text{ m} = 2.4\text{ m}$$

This freeboard is measured to Deck 3, the main vehicle deck, which serves as the vessel's deck line/freeboard deck.

Using the design draught as the summer load line mark and following the standard load line relationships outlined in the course notes [21], the remaining marks are calculated relative to the S mark. The standard load line relationships are as follows:

- Summer water mark for water of 1.025 t/m^3 density.
- Tropical water mark is 1/48 of summer load draft above the S mark.
- Fresh water mark is 1/48 of summer load draft above the S mark.
- Tropical freshwater mark is (T + F) above the S mark.
- Winter water mark is 1/48 of the summer load draft below the S mark.

- Winter North Atlantic water mark is placed 50 mm below the W mark (since vessel floodable length is less than 100 m, per Section 4.6).

Table 15: Freeboard Water Marks

Mark	Description	Offset from S [m]	Draft [m]
TF	Tropical Fresh	+0.171	4.271
F	Fresh	+0.085	4.185
T	Tropical	+0.085	4.185
S	Summer	-	4.1
W	Winter	-0.085	4.015
WNA	Winer North Atlantic	-0.135	3.965

These marks establish the range of permissible loaded drafts across seasonal operating conditions. Since the vessel's route lies within sheltered to moderately exposed coastal waters, the Summer and Tropical marks are expected to govern typical year-round operation, while the Winter and WNA marks provide additional reserve freeboard for the more severe wind and wave conditions described for the Satellite Channel during winter months.

4.7 Intact Stability

An intact stability assessment was completed on the vessel to validate its performance under a representative loading condition. The IMO MSC.267(85) Passenger Vessel Standards, from the International Code on Intact Stability (2008), Part A, §3.1, were selected as the governing criteria, as they most closely align with the vessel's requirements as a passenger ferry.

It should be noted that the severe wind and rolling criterion was not evaluated as part of this assessment, as it requires a defined above-water silhouette to calculate windage area, which has not yet been incorporated into the vessel model. As a result, this criterion appears as "Failed" in the results and should not be interpreted as a genuine stability failure.

Conservative assumptions were applied throughout to maintain a margin of safety in the results. Ballast tanks were set to 40% fill, while fuel and freshwater tanks were set to 98% fill. All weight and tank tonnages correspond to the values previously established in Section 4.4. The loading condition used for this assessment is summarized in Figure 15.

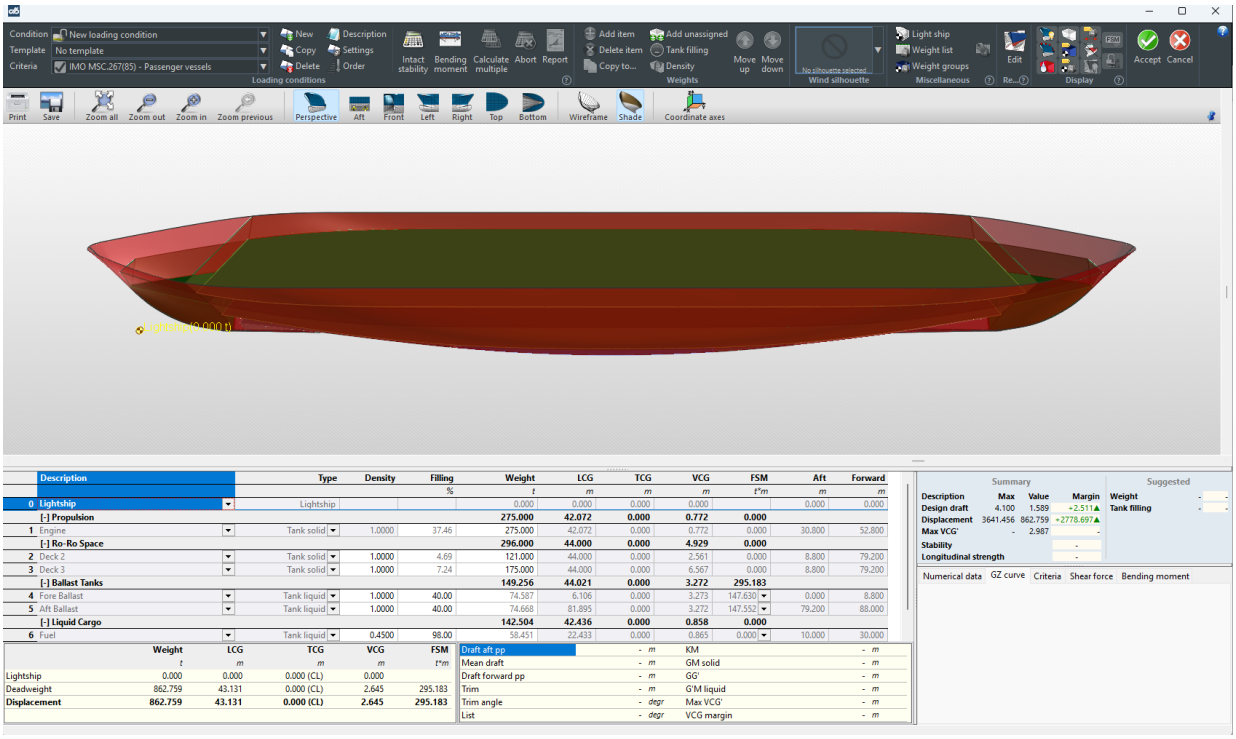


Figure 15: Intact Stability Loading Conditions

With the loading condition defined, the corresponding stability (GZ) curve was generated, shown in Figure 16.

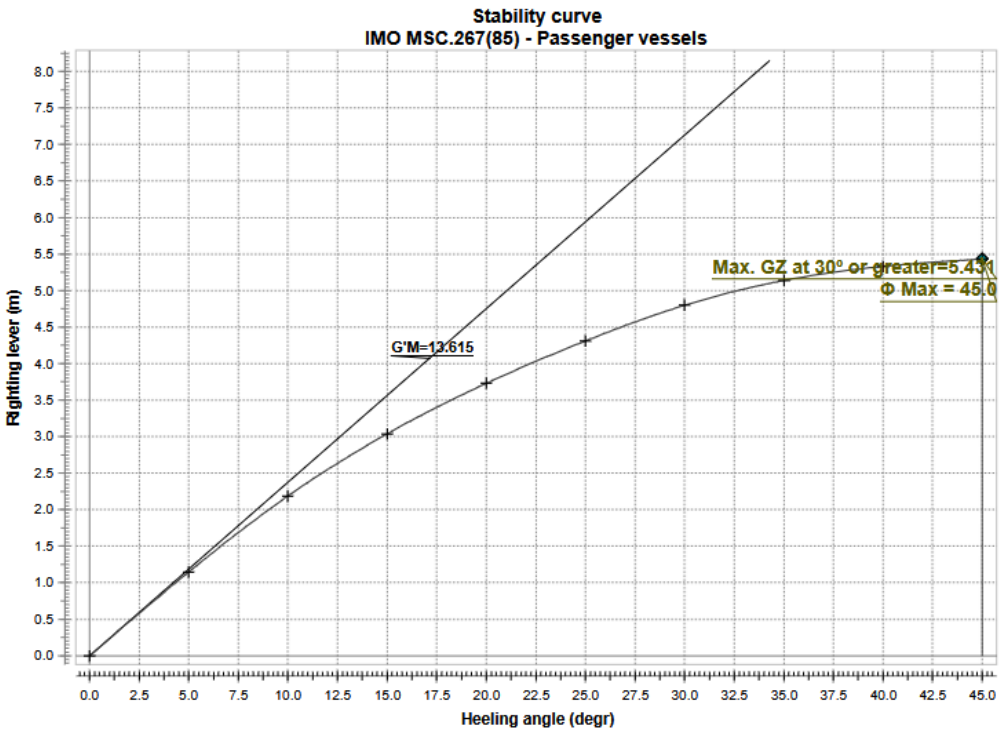


Figure 16: Intact Stability Curve

The resulting criteria are summarized in Figure 17. The vessel complies with all evaluated criteria, including initial metacentric height, maximum righting arm, angle of maximum righting arm, and area under the GZ curve. As noted above, the weather criterion is excluded from this evaluation and should be addressed in further design phases. The complete Intact Stability report can be found in Appendix F.

IMO MSC.267(85) - Passenger vessels

International Code on Intact Stability (2008), Part A, §3.1

Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	**1.4724 mrad**	>=	**0.0550 mrad**	**Complies**
Lower angle of range	0.0 degr			
Upper angle of range	30.0 degr			
Area 0° - 40°	**2.3648 mrad**	>=	**0.0900 mrad**	**Complies**
Lower angle of range	0.0 degr			
Upper angle of range	40.0 degr			
Area 30° - 40°	**0.8924 mrad**	>=	**0.0300 mrad**	**Complies**
Lower angle of range	30.0 degr			
Upper angle of range	40.0 degr			
Max. GZ at 30° or greater	**5.431 m**	>=	**0.200 m**	**Complies**
Lower angle of range	30.0 degr			
Upper angle of range	90.0 degr			
Angle of max GZ	**45.0 degr**	>=	**25.0 degr**	**Complies**
Initial metacentric height	**13.615 m**	>=	**0.150 m**	**Complies**
Severe wind and rolling criterion (weather criterion)				**Fails**
No silhouette selected				
Wind speed	50.54 kn			
Wind pressure	51.37 kg/m²			
Wind area	0.00 m²			
Steady wind lever	0.000 m			
Deck immersion angle	Could not be calculated! degr			
Wind gust lever	0.000 m			
Ratio of areaA/areaB	0.000	<=	1.000	Complies
Maximum allowed static heeling angle	0.0 degr	<=	16.0 degr	Complies
Max allowed ratio static angle/deck immersion angle	Could not be calculated!			Complies
Angle of equilibrium due to passenger crowding	**0.0 degr**	<	**10.0 degr**	**Complies**
Calculated heeling moment (upright)	0.000 t*m			
Turning angle	**0.0 degr**	<	**10.0 degr**	**Complies**
Calculated heeling moment (upright)	0.000 t*m			

The condition does NOT comply with the stability criteria

Figure 17: Intact Stability Results

5. Superstructure

Lastly, a superstructure was designed for the vessel, which included the enclosed passenger cabin (Deck 4), an open sun deck (Deck 5), forward and aft wheelhouses to support double-ended operation, and a smokestack to vent LNG/diesel engine exhaust. Footprint dimensions for each element are consistent with those originally established in Table 4 (Section 2.2). The following image represents the current proposed superstructure design for the Malahat Express.

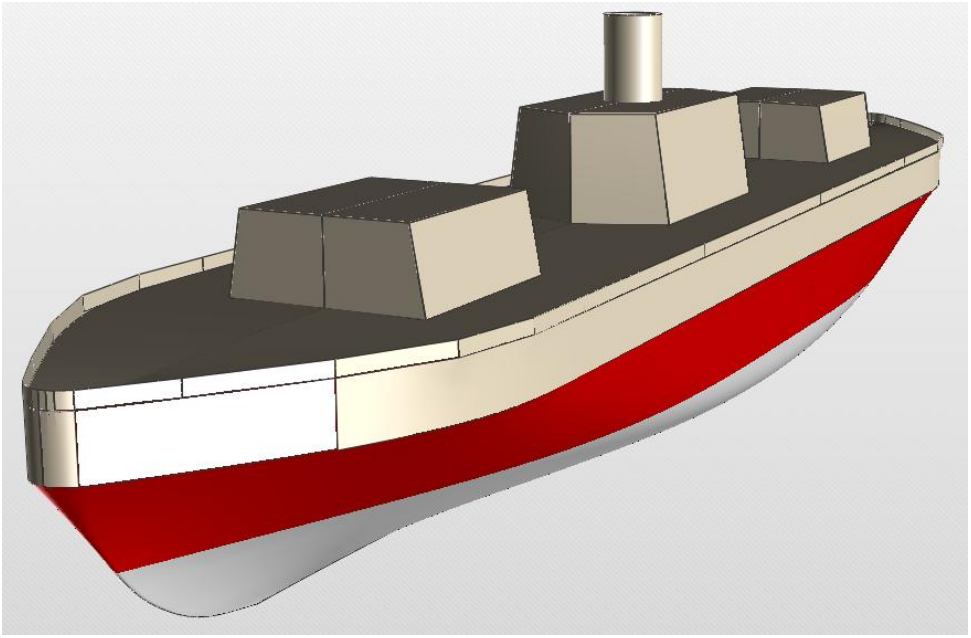


Figure 18: Malahat Express Perspective View

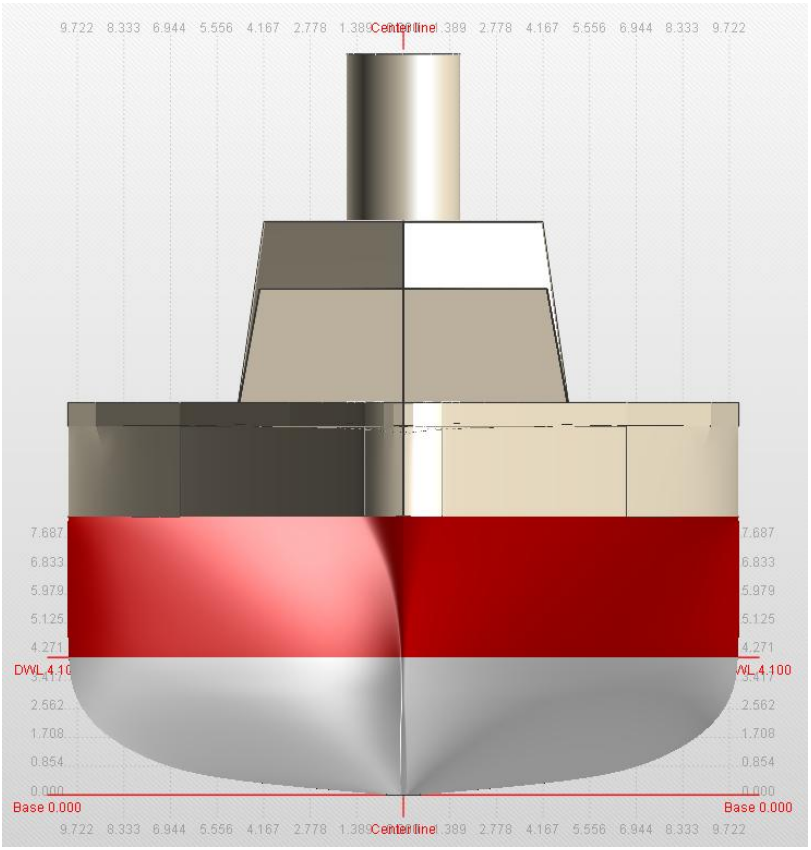


Figure 19: Malahat Express (Looking to Bow)

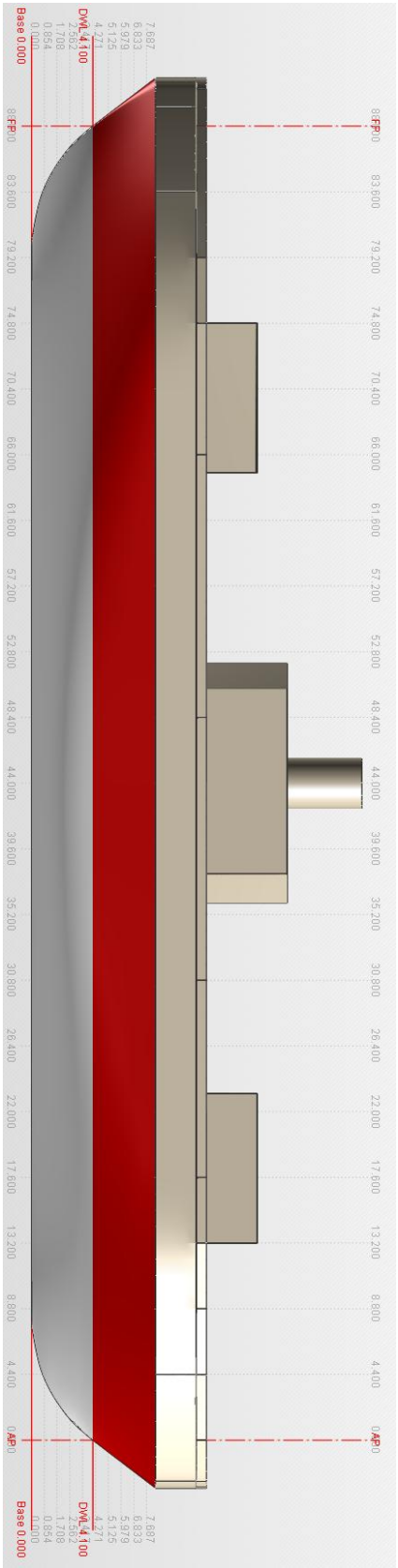


Figure 20: Malahat Express (Looking to Port)

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Appendix A: Bonjean Curve Data

Bonjean curves

Height	Area	VCG	Height	Area	VCG	Height	Area	VCG
m	m^2	m	m	m^2	m	m	m^2	m
Station 0.000								
4.119	0.00	0.000	**4.500**	0.12	4.352	**5.000**	0.51	4.680
5.500	1.20	5.020	**6.000**	2.24	5.366	**6.500**	3.67	5.714
Station 4.400								
0.751	0.00	0.751	**1.000**	0.07	0.893	**1.500**	0.34	1.197
2.000	0.81	1.531	**2.500**	1.54	1.877	**3.000**	2.55	2.229
3.500	3.89	2.584	**4.000**	5.61	2.946	**4.500**	7.78	3.311
5.000	10.41	3.676	**5.500**	13.53	4.041	**6.000**	17.18	4.405
6.500	21.36	4.767						
Station 8.800								
0.500	0.27	0.303	**1.000**	0.91	0.637	**1.500**	1.97	0.975
2.000	3.48	1.316	**2.500**	5.46	1.659	**3.000**	7.95	2.003
3.500	10.97	2.348	**4.000**	14.54	2.694	**4.500**	18.69	3.040
5.000	23.41	3.386	**5.500**	28.72	3.732	**6.000**	34.62	4.076
6.500	41.11	4.420						
Station 13.200								
0.000	0.00	0.000	**0.500**	0.59	0.298	**1.000**	1.87	0.624
1.500	3.87	0.954	**2.000**	6.57	1.287	**2.500**	9.99	1.619
3.000	14.10	1.951	**3.500**	18.87	2.280	**4.000**	24.26	2.608
4.500	30.26	2.934	**5.000**	36.84	3.259	**5.500**	43.97	3.583
6.000	51.65	3.905	**6.500**	59.84	4.227			
Station 17.600								
0.000	0.00	0.000	**0.500**	1.01	0.294	**1.000**	3.09	0.615
1.500	6.19	0.940	**2.000**	10.24	1.264	**2.500**	15.15	1.586
3.000	20.85	1.905	**3.500**	27.25	2.222	**4.000**	34.27	2.536
4.500	41.84	2.847	**5.000**	49.89	3.154	**5.500**	58.39	3.460
6.000	67.28	3.762	**6.500**	76.50	4.063			
Station 22.000								
0.000	0.00	0.000	**0.500**	1.38	0.301	**1.000**	4.35	0.622

Figure 21: Bonjean Curve Data (1 of 4)

Height	Area	VCG	Height	Area	VCG	Height	Area	VCG
m	m²	m	m	m²	m	m	m²	m
1.500	8.63	0.939	**2.000**	14.00	1.253	**2.500**	20.28	1.563
3.000	27.34	1.871	**3.500**	35.05	2.175	**4.000**	43.32	2.476
4.500	52.04	2.774	**5.000**	61.11	3.067	**5.500**	70.46	3.357
6.000	80.04	3.644	**6.500**	89.79	3.927			
Station 26.400								
0.000	0.00	0.000	**0.500**	1.79	0.308	**1.000**	5.73	0.624
1.500	11.16	0.934	**2.000**	17.71	1.238	**2.500**	25.16	1.539
3.000	33.31	1.836	**3.500**	42.03	2.130	**4.000**	51.16	2.419
4.500	60.59	2.704	**5.000**	70.24	2.985	**5.500**	80.02	3.263
6.000	89.90	3.536	**6.500**	99.84	3.806			
Station 30.800								
0.000	0.00	0.000	**0.500**	2.24	0.311	**1.000**	7.13	0.624
1.500	13.60	0.925	**2.000**	21.14	1.221	**2.500**	29.48	1.513
3.000	38.41	1.801	**3.500**	47.76	2.085	**4.000**	57.39	2.365
4.500	67.18	2.639	**5.000**	77.07	2.910	**5.500**	87.01	3.178
6.000	96.99	3.442	**6.500**	106.98	3.705			
Station 35.200								
0.000	0.00	0.000	**0.500**	2.68	0.314	**1.000**	8.40	0.621
1.500	15.69	0.916	**2.000**	23.95	1.205	**2.500**	32.90	1.490
3.000	42.30	1.770	**3.500**	52.00	2.047	**4.000**	61.87	2.318
4.500	71.82	2.586	**5.000**	81.81	2.850	**5.500**	91.81	3.112
6.000	101.82	3.371	**6.500**	111.83	3.629			
Station 39.600								
0.000	0.00	0.000	**0.500**	3.02	0.315	**1.000**	9.33	0.618
1.500	17.10	0.908	**2.000**	25.74	1.192	**2.500**	34.96	1.471
3.000	44.56	1.747	**3.500**	54.38	2.019	**4.000**	64.32	2.286
4.500	74.30	2.550	**5.000**	84.30	2.811	**5.500**	94.31	3.070
6.000	104.33	3.327	**6.500**	114.34	3.583			
Station 44.000								
0.000	0.00	0.000	**0.500**	3.18	0.315	**1.000**	9.74	0.616
1.500	17.70	0.903	**2.000**	26.47	1.185	**2.500**	35.79	1.463

Figure 22: Bonjean Curve Data (2 of 4)

Height	Area	VCG	Height	Area	VCG	Height	Area	VCG
m	m²	m	m	m²	m	m	m²	m
3.000	45.45	1.737	**3.500**	55.30	2.007	**4.000**	65.25	2.272
4.500	75.23	2.535	**5.000**	85.23	2.795	**5.500**	95.23	3.053
6.000	105.24	3.309	**6.500**	115.25	3.564			
Station 48.400								
0.000	0.00	0.000	**0.500**	3.02	0.315	**1.000**	9.33	0.618
1.500	17.10	0.908	**2.000**	25.74	1.192	**2.500**	34.96	1.471
3.000	44.56	1.747	**3.500**	54.38	2.019	**4.000**	64.32	2.286
4.500	74.30	2.550	**5.000**	84.30	2.811	**5.500**	94.31	3.070
6.000	104.33	3.327	**6.500**	114.34	3.583			
Station 52.800								
0.000	0.00	0.000	**0.500**	2.68	0.314	**1.000**	8.40	0.621
1.500	15.69	0.916	**2.000**	23.95	1.205	**2.500**	32.90	1.490
3.000	42.30	1.770	**3.500**	52.00	2.047	**4.000**	61.87	2.318
4.500	71.82	2.586	**5.000**	81.81	2.850	**5.500**	91.81	3.112
6.000	101.82	3.371	**6.500**	111.83	3.629			
Station 57.200								
0.000	0.00	0.000	**0.500**	2.24	0.311	**1.000**	7.13	0.624
1.500	13.60	0.925	**2.000**	21.14	1.221	**2.500**	29.48	1.513
3.000	38.41	1.801	**3.500**	47.76	2.085	**4.000**	57.39	2.365
4.500	67.18	2.639	**5.000**	77.07	2.910	**5.500**	87.01	3.178
6.000	96.99	3.442	**6.500**	106.98	3.705			
Station 61.600								
0.000	0.00	0.000	**0.500**	1.79	0.308	**1.000**	5.73	0.624
1.500	11.16	0.934	**2.000**	17.71	1.238	**2.500**	25.16	1.539
3.000	33.31	1.836	**3.500**	42.03	2.130	**4.000**	51.16	2.419
4.500	60.59	2.704	**5.000**	70.24	2.985	**5.500**	80.02	3.263
6.000	89.90	3.536	**6.500**	99.84	3.806			
Station 66.000								
0.000	0.00	0.000	**0.500**	1.38	0.301	**1.000**	4.35	0.622
1.500	8.63	0.939	**2.000**	14.00	1.253	**2.500**	20.28	1.563
3.000	27.34	1.871	**3.500**	35.05	2.175	**4.000**	43.32	2.476

Figure 23: Bonjean Curve Data (3 of 4)

Height	Area	VCG	Height	Area	VCG	Height	Area	VCG
m	m²	m	m	m²	m	m	m²	m
4.500	52.04	2.774	**5.000**	61.11	3.067	**5.500**	70.46	3.357
6.000	80.04	3.644	**6.500**	89.79	3.927			
Station 70.400								
0.000	0.00	0.000	**0.500**	1.01	0.294	**1.000**	3.09	0.615
1.500	6.19	0.940	**2.000**	10.24	1.264	**2.500**	15.15	1.586
3.000	20.85	1.905	**3.500**	27.25	2.222	**4.000**	34.27	2.536
4.500	41.84	2.847	**5.000**	49.89	3.154	**5.500**	58.39	3.460
6.000	67.28	3.762	**6.500**	76.50	4.063			
Station 74.800								
0.000	0.00	0.000	**0.500**	0.59	0.298	**1.000**	1.87	0.624
1.500	3.87	0.954	**2.000**	6.57	1.287	**2.500**	9.99	1.619
3.000	14.10	1.951	**3.500**	18.87	2.280	**4.000**	24.26	2.608
4.500	30.26	2.934	**5.000**	36.84	3.259	**5.500**	43.97	3.583
6.000	51.65	3.905	**6.500**	59.84	4.227			
Station 79.200								
0.500	0.27	0.304	**1.000**	0.91	0.638	**1.500**	1.97	0.976
2.000	3.48	1.317	**2.500**	5.46	1.660	**3.000**	7.95	2.004
3.500	10.97	2.349	**4.000**	14.54	2.694	**4.500**	18.69	3.041
5.000	23.41	3.387	**5.500**	28.72	3.732	**6.000**	34.62	4.076
6.500	41.11	4.420						
Station 83.600								
0.751	0.00	0.751	**1.000**	0.07	0.893	**1.500**	0.34	1.197
2.000	0.81	1.531	**2.500**	1.54	1.877	**3.000**	2.55	2.229
3.500	3.89	2.584	**4.000**	5.61	2.946	**4.500**	7.78	3.311
5.000	10.41	3.676	**5.500**	13.53	4.041	**6.000**	17.18	4.405
6.500	21.36	4.767						
Station 88.000								
4.119	0.00	0.000	**4.500**	0.12	4.352	**5.000**	0.51	4.680
5.500	1.20	5.020	**6.000**	2.24	5.366	**6.500**	3.67	5.714

Figure 24: Bonjean Curve Data (4 of 4)

Appendix B: Overall Lines Plan

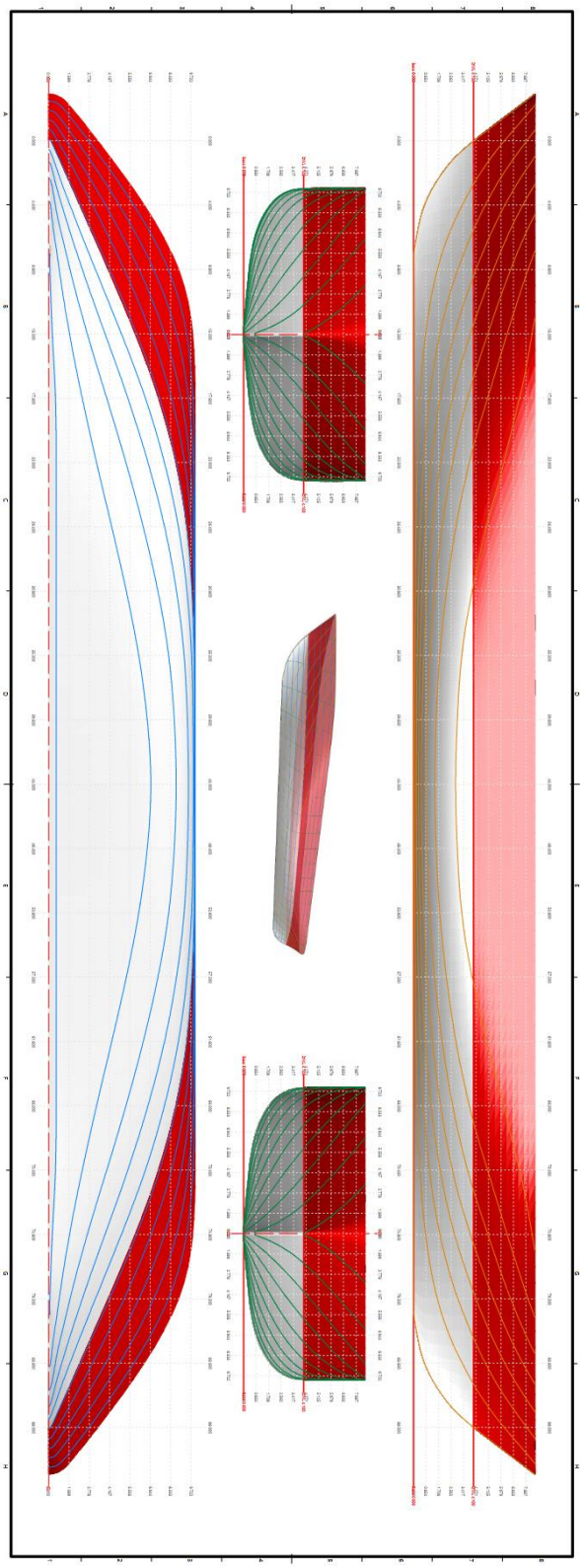


Figure 25: Lines Plan

Appendix C: DELFTship Hydrostatics Report

Design hydrostatics report

Malahat Express

Designer	Jackson Hamel
Created by	Jackson Hamel
Comment	
Filename	Malahat Express.fbm

Design length	88.000 m	**Midship location**	44.000 m
Length of buoyancy model	94.405 m	**Water density**	1.0250
Design beam	20.000 m	**Mean shell thickness**	0.0000 m
Maximum beam	20.033 m	**Appendage coefficient**	1.0000
Design draft	4.100 m		

Volume properties		Waterplane properties	
Moulded volume	3552.149 m^3	Length on waterline	87.969 m
Total displaced volume	3552.149 m^3	Beam on waterline	19.953 m
Displacement	3640.952 t	Entrance angle	57.0 degr
Block coefficient	0.4936	Waterplane area	1269.28 m^2
Prismatic coefficient	0.6005	Waterplane coefficient	0.7231
Vert. prismatic coefficient	0.6826	Waterplane center of floatation	44.000 m
Wetted surface area	1533.55 m^2	Transverse moment of inertia	32443.814 m^4
Longitudinal center of buoyancy	44.002 m	Longitudinal moment of inertia	510683.858 m^4
Longitudinal center of buoyancy	0.003 %		
Vertical center of buoyancy	2.466 m		
Total length of submerged body	87.969 m		
Total beam of submerged body	19.953 m		

Midship properties		Initial stability	
Midship section area	67.25 m^2	Transverse metacentric height	11.600 m
Midship coefficient	0.8220	Longitudinal metacentric height	146.234 m

Lateral plane	
Lateral area	340.02 m^2
Longitudinal center of lateral resistance	44.000 m
Vertical center of lateral resistance	2.104 m

The following layer properties are calculated for both sides of the ship

Location	Area	Thickness	Weight	LCG	TCG	VCG
	m^2	m	t	m	m	m
Hull	2477.60	0.050	953.878	44.001	0.000 (CL)	3.300
Superstructure	4446.57	0.050	1711.930	43.987	0.000 (CL)	11.787
Total	**6924.17**		**2665.807**	**43.992**	**0.000 (CL)**	**8.750**

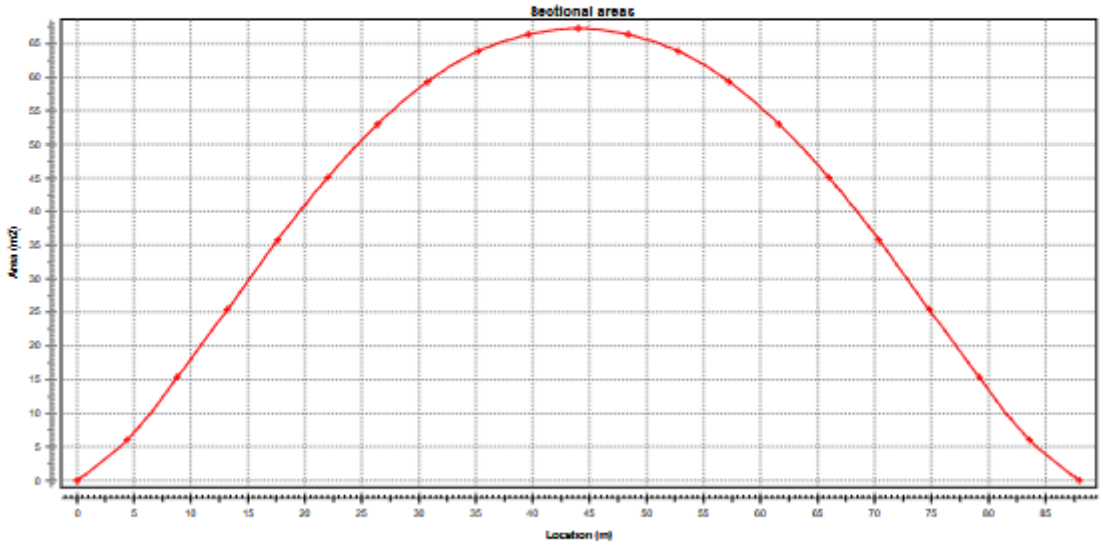
Sectional areas

Location	Area	Location	Area	Location	Area	Location	Area	Location	Area
m	m^2	m	m^2	m	m^2	m	m^2	m	m^2
0.000	0.00	22.000	45.03	44.000	67.25	66.000	45.03	88.000	0.00
4.400	6.01	26.400	53.02	48.400	66.31	70.400	35.74		

Figure 26: Design Hydrostatics Report (1 of 2)

Sectional areas

Location	Area	Location	Area	Location	Area	Location	Area	Location	Area
m	m²	m	m²	m	m²	m	m²	m	m²
8.800	15.32	30.800	59.33	52.800	63.85	74.800	25.41		
13.200	25.41	35.200	63.85	57.200	59.33	79.200	15.33		
17.600	35.74	39.600	66.31	61.600	53.02	83.600	6.01		



NOTE 1: Draft (and all other vertical heights) is measured from base Z=0.000
NOTE 2: All calculated coefficients based on actual dimensions of submerged body.

Figure 27: Design Hydrostatics Report (2 of 2)

Appendix D: Floodable Length Calculations

Floodable length calculation.

Input data		
Displacement	3637.505	t
LCG	44.010	m
VCG	0.000	m
Permeability	0.950	

Location	Floodable length	Flooded from	Flooded to	Flooded volume	Flooded LCG	Draft	Trim
m	m	m	m	m^3	m	m	m
10.740	27.884	-3.202	24.682	1489.840	14.918	5.041	-6.074
10.798	27.614	-3.009	24.605	1471.605	14.893	5.032	-6.017
11.798	25.971	-1.187	24.783	1494.765	15.031	5.045	-6.066
12.798	24.610	0.493	25.103	1509.871	15.373	5.058	-6.043
13.798	23.543	2.027	25.569	1521.845	15.893	5.071	-5.969
14.798	22.701	3.448	26.148	1531.578	16.538	5.084	-5.860
15.798	22.126	4.735	26.861	1553.304	17.276	5.106	-5.765
16.798	21.751	5.923	27.673	1584.467	18.075	5.133	-5.681
17.798	21.538	7.029	28.567	1623.701	18.916	5.166	-5.604
18.798	21.457	8.070	29.526	1669.989	19.787	5.203	-5.531
19.798	21.489	9.053	30.543	1722.736	20.680	5.245	-5.459
20.798	21.618	9.989	31.607	1781.333	21.590	5.291	-5.386
21.798	21.837	10.880	32.716	1845.944	22.512	5.341	-5.309
22.798	22.138	11.729	33.867	1916.623	23.444	5.395	-5.229
23.798	22.518	12.539	35.057	1993.453	24.383	5.454	-5.142
24.798	22.969	13.313	36.283	2076.392	25.328	5.518	-5.049
25.798	23.492	14.052	37.544	2165.625	26.278	5.585	-4.946
26.798	24.080	14.758	38.838	2261.316	27.231	5.658	-4.833
27.798	24.734	15.431	40.165	2364.018	28.187	5.735	-4.710
28.798	25.453	16.071	41.525	2474.150	29.145	5.818	-4.573
29.798	26.237	16.679	42.917	2592.632	30.106	5.906	-4.425
30.798	27.087	17.255	44.341	2720.431	31.068	6.000	-4.262
31.798	28.005	17.796	45.800	2858.513	32.031	6.102	-4.085
32.798	28.989	18.303	47.293	3007.651	32.995	6.210	-3.891
33.798	29.819	18.889	48.707	3132.076	33.963	6.303	-3.648
34.798	30.633	19.481	50.115	3254.900	34.933	6.394	-3.382
35.798	30.884	20.356	51.240	3283.689	35.911	6.424	-3.035
36.798	30.865	21.365	52.231	3268.347	36.893	6.424	-2.656
37.798	30.852	22.372	53.224	3255.302	37.877	6.424	-2.281
38.798	30.833	23.382	54.214	3242.910	38.863	6.423	-1.908
39.798	30.824	24.386	55.210	3233.873	39.849	6.423	-1.538
40.798	30.817	25.390	56.206	3226.753	40.836	6.423	-1.171
41.798	30.811	26.393	57.203	3221.257	41.824	6.423	-0.805
42.798	30.811	27.393	58.203	3218.281	42.812	6.423	-0.438
43.798	30.818	28.389	59.207	3218.237	43.800	6.424	-0.073
44.798	30.811	29.393	60.203	3217.531	44.788	6.423	0.292
45.798	30.812	30.392	61.204	3219.976	45.776	6.423	0.657

Figure 28: Floodable Length Report (1 of 2)

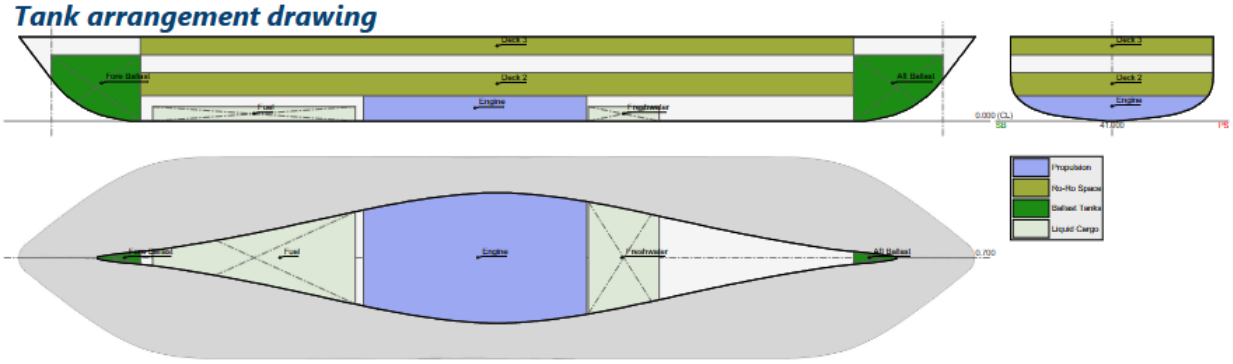
Location m	Floodable length m	Flooded from m	Flooded to m	Flooded volume m^3	Flooded LCG m	Draft m	Trim m
46.798	30.816	31.390	62.206	3224.336	46.765	6.423	1.025
47.798	30.822	32.387	63.209	3230.715	47.752	6.423	1.392
48.798	30.827	33.385	64.211	3238.707	48.738	6.423	1.760
49.798	30.844	34.376	65.220	3250.058	49.724	6.424	2.131
50.798	30.857	35.369	66.227	3262.311	50.708	6.424	2.505
51.798	30.875	36.360	67.236	3276.767	51.691	6.424	2.883
52.798	30.903	37.346	68.250	3294.392	52.672	6.425	3.263
53.798	30.132	38.732	68.864	3178.466	53.644	6.338	3.542
54.798	29.383	40.107	69.489	3066.959	54.613	6.254	3.805
55.798	28.392	41.602	69.994	2916.346	55.578	6.144	4.009
56.798	27.448	43.074	70.522	2774.076	56.542	6.040	4.193
57.798	26.573	44.512	71.084	2642.472	57.504	5.943	4.361
58.798	25.763	45.917	71.679	2520.317	58.465	5.852	4.515
59.798	25.018	47.289	72.307	2406.906	59.424	5.768	4.656
60.798	24.337	48.630	72.966	2301.174	60.381	5.688	4.785
61.798	23.722	49.937	73.659	2202.813	61.336	5.614	4.902
62.798	23.176	51.210	74.386	2111.268	62.287	5.545	5.008
63.798	22.696	52.450	75.146	2025.978	63.234	5.480	5.106
64.798	22.288	53.654	75.942	1946.895	64.177	5.419	5.195
65.798	21.955	54.820	76.776	1873.805	65.112	5.363	5.278
66.798	21.701	55.947	77.649	1806.631	66.039	5.311	5.356
67.798	21.533	57.032	78.564	1745.385	66.954	5.263	5.430
68.798	21.458	58.069	79.527	1690.069	67.855	5.219	5.502
69.798	21.490	59.053	80.543	1641.184	68.737	5.180	5.575
70.798	21.645	59.976	81.620	1599.057	69.592	5.146	5.650
71.798	21.948	60.824	82.772	1564.545	70.411	5.116	5.731
72.798	22.432	61.582	84.014	1538.741	71.178	5.092	5.821
73.798	23.150	62.223	85.373	1523.567	71.868	5.075	5.924
74.798	24.149	62.723	86.873	1517.711	72.445	5.065	6.030
75.798	25.337	63.130	88.466	1498.660	72.874	5.049	6.059
76.798	26.951	63.323	90.273	1489.384	73.084	5.041	6.074
77.271	27.863	63.339	91.202	1488.560	73.103	5.041	6.075

Figure 29: Floodable Length Report (2 of 2)

Appendix E: Tank Capacities Report

Tank and compartment layout

Tank arrangement drawing



Tanks and compartments

Propulsion

Number	Tank description	Short name	Density	Moulded volume	Volume	Weight	LCG	TCG	VCG	Max FSM
				m³	m³	t	m	m	m	t*m
	Engine		1.0000	749.085	734.103	734.103	41.987	0.000 (CL)	1.478	11649.626
Total					**734.103**	**734.103**	**41.987**	**0.000 (CL)**	**1.478**	**11649.626**

Ro-Ro Space

Number	Tank description	Short name	Density	Moulded volume	Volume	Weight	LCG	TCG	VCG	Max FSM
				m³	m³	t	m	m	m	t*m
	Deck 2		1.0000	2632.856	2580.199	2580.199	44.000	0.000 (CL)	3.691	34545.807
	Deck 3		1.0000	2465.986	2416.666	2416.666	43.998	0.000 (CL)	7.405	45070.612
Total					**4996.865**	**4996.865**	**43.999**	**0.000 (CL)**	**5.487**	**79616.419**

Ballast Tanks

Number	Tank description	Short name	Density	Moulded volume	Volume	Weight	LCG	TCG	VCG	Max FSM
				m³	m³	t	m	m	m	t*m

Figure 30: Tank Capacities Report (1 of 2)

	Fore Ballast		1.0000	190.273	186.468	186.468	5.700	0.000 (CL)	4.689	644.638
	Aft Ballast		1.0000	190.481	186.671	186.671	82.302	0.000 (CL)	4.688	644.646
Total					**373.139**	**373.139**	**44.022**	**0.000 (CL)**	**4.688**	**1289.284**

Liquid Cargo

Number	Tank description	Short name	Density	Moulded volume	Volume	Weight	LCG	TCG	VCG	Max FSM
				m^3	m^3	t	m	m	m	t*m
	Fuel		0.4500	135.248	132.543	59.644	22.429	0.000 (CL)	0.876	564.117
	Freshwater		1.0000	87.518	85.768	85.768	56.347	0.000 (CL)	0.864	1550.711
Total					**218.310**	**145.412**	**42.435**	**0.000 (CL)**	**0.869**	**2114.828**

Figure 31: Tank Capacities Report (2 of 2)

Appendix F: Intact Stability

New loading condition

Malahat Express

Designer	Jackson Hamel
Created by	Jackson Hamel
Comment	
Filename	Malahat Express.fbm

Design length	88.000 m	**Midship location**	44.000 m
Length of buoyancy model	94.405 m	**Water density**	1.0250
Design beam	20.000 m	**Mean shell thickness**	0.0000 m
Maximum beam	20.033 m	**Appendage coefficient**	1.0000
Design draft	4.100 m		

Calculation settings

Heel to	:	**Side with least stability**
Center of gravity of tanks containing liquids	:	**Actual COG**

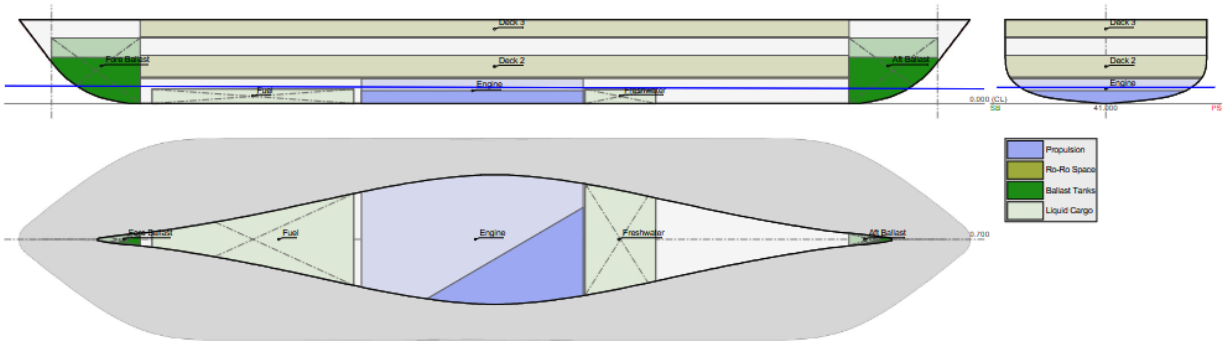


Figure 32: Intact Stability Report (1 of 4)

Hydrostatic particulars

List	0.0 (CL) degr	**VCG'**	2.987 m
Draft aft pp	1.726 m	**Max VCG'**	0.000 m
Mean moulded draft	1.589 m	**GM solid**	13.957 m
Draft forward pp	1.452 m	**G'M liquid**	13.615 m
Trim	-0.275 m	**Immersion rate**	8.510 t/cm
LCF	43.566 m	**MCT**	27.818 t*m
LCB	43.117 m	**Lwl**	82.472 m
KM	16.602 m	**Bwl**	17.110 m
VCG	2.645 m	**LCB %**	-1.07 %
GG'	0.342 m		

Description	Density	Filling	Weight	LCG	TCG	VCG	FSM
		%	t	m	m	m	t*m
Propulsion							
Engine	1.0000	37.5	275.000	42.072	0.000 (CL)	0.772	0.000
Ro-Ro Space							
Deck 2	1.0000	4.7	121.000	44.000	0.000 (CL)	2.561	0.000
Deck 3	1.0000	7.2	175.000	44.000	0.000 (CL)	6.567	0.000
Totals for Ro-Ro Space			**296.000**	**44.000**	**0.000 (CL)**	**4.929**	
Ballast Tanks							
Fore Ballast	1.0000	40.0	74.587	6.106	0.000 (CL)	3.273	147.630
Aft Ballast	1.0000	40.0	74.668	81.895	0.000 (CL)	3.272	147.552
Totals for Ballast Tanks			**149.256**	**44.021**	**0.000 (CL)**	**3.272**	**295.183**
Liquid Cargo							
Fuel	0.4500	98.0	58.451	22.433	0.000 (CL)	0.865	0.000
Freshwater	1.0000	98.0	84.052	56.347	0.000 (CL)	0.853	0.000
Totals for Liquid Cargo			**142.504**	**42.436**	**0.000 (CL)**	**0.858**	**0.000**
Lightship			**0.000**	**0.000**	**0.000 (CL)**	**0.000**	
Deadweight			**862.759**	**43.131**	**0.000 (CL)**	**2.645**	**295.183**
Displacement			**862.759**	**43.131**	**0.000 (CL)**	**2.645**	**295.183**

Figure 33: Intact Stability Report (2 of 4)

Righting levers

Heeling angle	Draft	Trim	Displacement	KN sin(ø)	VCG sin(ø)	GG' sin(ø)	TCG cos(ø)	GZ	Area
degr	m	m	t	m	m	m	m	m	mrad
0.0 (CL)	1.589	-0.275	862.752	0.000	0.000	0.000	0.000	0.000	0.0000
5.0 (PS)	1.544	-0.277	862.755	1.426	0.231	0.048	0.000	1.147	0.0504
10.0 (PS)	1.408	-0.286	862.755	2.726	0.459	0.081	0.000	2.186	0.1971
15.0 (PS)	1.184	-0.302	862.756	3.840	0.685	0.118	0.000	3.038	0.4264
20.0 (PS)	0.879	-0.324	862.754	4.795	0.905	0.161	0.000	3.729	0.7226
25.0 (PS)	0.491	-0.356	862.754	5.629	1.118	0.200	0.000	4.311	1.0740
30.0 (PS)	0.005	-0.424	862.752	6.352	1.323	0.230	0.000	4.799	1.4724
35.0 (PS)	-0.618	-0.535	862.757	6.907	1.517	0.253	0.000	5.137	1.9071
40.0 (PS)	-1.416	-0.610	862.751	7.300	1.700	0.269	0.000	5.330	2.3648
45.0 (PS)	-2.424	-0.664	862.755	7.583	1.870	0.281	0.000	5.431	2.8347

Intact stability Malahat Express

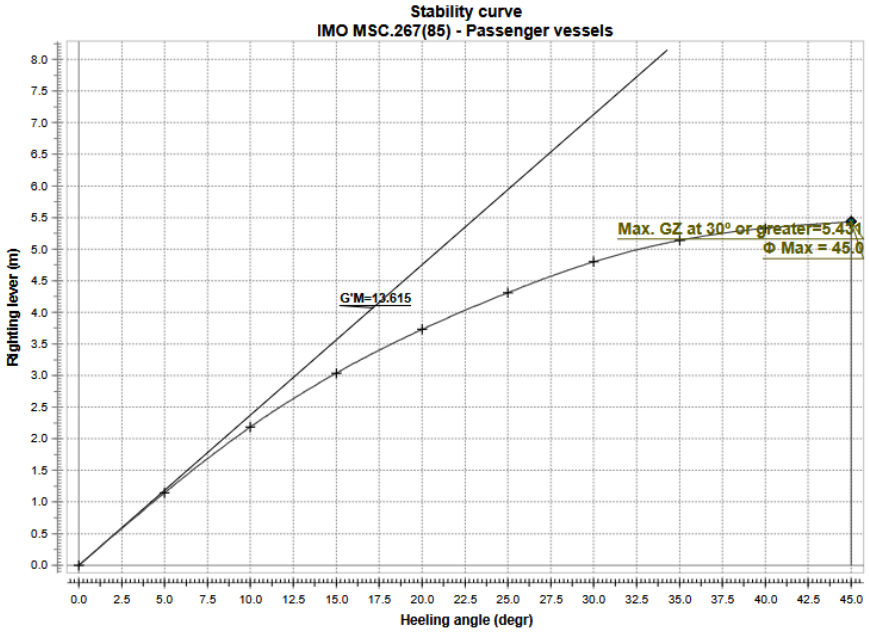


Figure 34: Intact Stability Report (3 of 4)

IMO MSC.267(85) - Passenger vessels

International Code on Intact Stability (2008), Part A, §3.1

Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	**1.4724 mrad**	>=	**0.0550 mrad**	**Complies**
Lower angle of range	0.0 degr			
Upper angle of range	30.0 degr			
Area 0° - 40°	**2.3648 mrad**	>=	**0.0900 mrad**	**Complies**
Lower angle of range	0.0 degr			
Upper angle of range	40.0 degr			
Area 30° - 40°	**0.8924 mrad**	>=	**0.0300 mrad**	**Complies**
Lower angle of range	30.0 degr			
Upper angle of range	40.0 degr			
Max. GZ at 30° or greater	**5.431 m**	>=	**0.200 m**	**Complies**
Lower angle of range	30.0 degr			
Upper angle of range	90.0 degr			
Angle of max GZ	**45.0 degr**	>=	**25.0 degr**	**Complies**
Initial metacentric height	**13.615 m**	>=	**0.150 m**	**Complies**
Severe wind and rolling criterion (weather criterion)				**Fails**
No silhouette selected				
Wind speed	50.54 kn			
Wind pressure	51.37 kg/m^2			
Wind area	0.00 m^2			
Steady wind lever	0.000 m			
Deck immersion angle	Could not be calculated! degr			
Wind gust lever	0.000 m			
Ratio of areaA/areaB	0.000	<=	1.000	Complies
Maximum allowed static heeling angle	0.0 degr	<=	16.0 degr	Complies
Max allowed ratio static angle/deck immersion angle	Could not be calculated!			Complies
Angle of equilibrium due to passenger crowding	**0.0 degr**	<	**10.0 degr**	**Complies**
Calculated heeling moment (upright)	0.000 t*m			
Turning angle	**0.0 degr**	<	**10.0 degr**	**Complies**
Calculated heeling moment (upright)	0.000 t*m			

The condition does NOT comply with the stability criteria

Figure 35: Intact Stability Report (4 of 4)